21

The Kinetic Theory of Gases

CHAPTER OUTLINE

21.1 Molecular Model of an Ideal Gas

21.2 Molar Specific Heat of an Ideal Gas

21.3 The Equipartition of Energy

21.4 Adiabatic Processes for an Ideal Gas

21.5 Distribution of Molecular Speeds

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ21.1 Answer (c). The molecular mass of nitrogen (N_2, 28 u) is smaller than the molecular mass of oxygen (O_2, 32 u), and the rms speed of a gas is $(3RT/M)^{1/2}$. Since the rms speeds are the same, the temperature of nitrogen is smaller than the temperature of oxygen. The average kinetic energy is proportional to the molecular mass and the square of the rms speed ($K = \frac{1}{2}mv_{\text{rms}}^2$), so the average kinetic energy of nitrogen is smaller.

OQ21.2 Answer (d). The rms speed of molecules in the gas is $v_{\text{rms}} = \sqrt{3RT/M}$. Thus, the ratio of the final speed to the original speed would be

$$\frac{(v_{\text{rms}})_f}{(v_{\text{rms}})_0} = \frac{\sqrt{3RT_f/M}}{\sqrt{3RT_0/M}} = \sqrt{\frac{T_f}{T_0}} = \sqrt{\frac{600\text{ K}}{200\text{ K}}} = \sqrt{3}$$

OQ21.3 Answer (b). The gases are the same so they have the same molecular mass, M. If the two samples have the same density, then their ratios of number of moles to volume, n/V, are the same because their

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densities, $(nM)/V$, are the same. The pressures are the same; thus, their temperatures are the same:

$$PV = nRT \rightarrow p = \frac{n}{V}RT = \text{ constant } \rightarrow T = \text{ constant}$$

Therefore the rms speed of their molecules, $(3RT/M)^{1/2}$, is the same.

OQ21.4 (i) Answer (b). The volume of the balloon will decrease because the gas cools.

(ii) Answer (c). The pressure inside the balloon is nearly equal to the constant exterior atmospheric pressure. Snap the mouth of the balloon over an absolute pressure gauge to demonstrate this fact. Then from $PV = nRT$, volume must decrease in proportion to the absolute temperature. Call the process isobaric contraction.

OQ21.5 Answer (d). At 200 K, $\frac{1}{2}m_0 v_{\text{rms0}}^2 = \frac{3}{2}k_B T_0$. At the higher temperature,

$$\frac{1}{2}m_0\left(2v_{\text{rms0}}\right)^2 = \frac{3}{2}k_B T$$

Then $T = 4T_0 = 4(200 \text{ K}) = 800 \text{ K}$.

OQ21.6 Answer (c) > (a) > (b) > (d). The average vector velocity is zero in a sample macroscopically at rest. As adjacent equations in the text note, the asymmetric distribution of molecular speeds makes the average speed greater than the most probable speed, and the rms speed greater still. The most probable speed is $(2RT/M)^{1/2}$, the average speed is $(8RT/\pi M)^{1/2} \cong (2.55RT/M)^{1/2}$, and the rms speed is $(3RT/M)^{1/2}$.

OQ21.7 (i) Statements (a) and (e) are correct statements that describe the temperature increase of a gas.

(ii) Statement (f) is a correct statement but does not apply to the situation. Statement (b) is true if the molecules have any size at all, but molecular collisions with other molecules have nothing to do with the temperature increase.

(iii) Statements (c) and (d) are incorrect. The molecular collisions are perfectly elastic. Temperature is determined by how fast molecules are moving through space, not by anything going on inside a molecule.

OQ21.8 (i) Answer (b). Average molecular kinetic energy, $3kT/2$, increases by a factor of 3.

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(ii) Answer (c). The rms speed, $(3RT/M)^{1/2}$, increases by a factor of $\sqrt{3}$.

(iii) Answer (c). Average momentum change increases by $\sqrt{3}$: $\Delta p_{\text{avg}} = -2m_0 v_{\text{avg}}$.

(iv) Answer (c). Rate of collisions increases by a factor of $\sqrt{3}$: $\Delta t_{\text{avg}} = 2d / v_{\text{avg}}$.

(v) Answer (b). Pressure increases by a factor of 3. See Equation 21.15:

$$P = \frac{2}{3}\left(\frac{N_i}{V}\right)\left(\frac{1}{2}m_0\overline{v^2}\right) = \frac{2}{3}\left(\frac{N_i}{V}\right)\left(\overline{K}\right)$$

OQ21.9 Answer (c). The kinetic theory of gases assumes that the molecules do not interact with each other.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ21.1 As a parcel of air is pushed upward, it moves into a region of lower pressure, so it expands and does work on its surroundings. Its fund of internal energy drops, and so does its temperature. As mentioned in the question, the low thermal conductivity of air means that very little energy will be conducted by heat into the now-cool parcel from the denser but warmer air below it.

CQ21.2 A diatomic gas has more degrees of freedom—those of molecular vibration and rotation—than a monatomic gas. The energy content per mole is proportional to the number of degrees of freedom.

CQ21.3 Alcohol evaporates rapidly, so that high-speed molecules leave the liquid, reducing the average kinetic energy of the remaining molecules of the liquid and therefore reducing the temperature of the liquid. *Then*, because the alcohol is cool, energy transfers from the skin, reducing its temperature.

CQ21.4 As the balloon rises into the air, the air cannot be uniform in pressure because the lower layers support the weight of all the air above them. The rubber in a typical balloon is easy to stretch and stretches or contracts until interior and exterior pressures are nearly equal. So as the balloon rises it expands. This is an adiabatic expansion (see Section 21.4), with P decreasing as V increases (PV^{γ} = constant). If the rubber wall is very strong it will eventually contain the helium at higher pressure than the air outside but at the same density, so that the balloon will stop rising. More likely, the rubber will stretch and break, releasing the helium to keep rising and "boil out" of the Earth's atmosphere.

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CQ21.5 The dry air is more dense. Since the air and the water vapor are at the same temperature, the gases have the same average molecular kinetic energy. Imagine a controlled experiment in which equal-volume containers, one with humid air and one with dry air, are at the same pressure. The number of molecules must be the same for both containers ($PV = NkT$). The water molecule has a smaller molecular mass (18.0 u) than any of the gases that make up the air, so the humid air must have the smaller mass per unit volume.

CQ21.6 The helium must have the higher rms speed. According to Equation 21.22 for the rms speed, $(3RT/M)^{1/2}$, for the same temperature, the gas with the smaller mass per atom must have the higher average speed squared and thus the higher rms speed.

CQ21.7 The molecules of all different kinds collide with the walls of the container, so molecules of all different kinds exert partial pressures that contribute to the total pressure. The molecules can be so small that they collide with one another relatively rarely and each kind exerts partial pressure as if the other kinds of molecules were absent. If the molecules collide with one another often, the collisions exactly conserve momentum and so do not affect the net force on the walls. The partial pressure P_i of one of the gases can be expressed with Equation 21.15:

$$P_i = \frac{2}{3}\left(\frac{N_i}{V}\right)\left(\frac{1}{2}m_0\overline{v^2}\right) = \frac{2}{3}\left(\frac{N_i}{V}\right)\left(\overline{K}\right)$$

where N_i is the number of molecules of the ith gas and $\overline{K}$ is the average kinetic energy of the molecules. Let us add up these pressures for all the gases in the container:

$$P = \sum_i P_i = \sum_i \frac{2}{3}\left(\frac{N_i}{V}\right)\left(\overline{K}\right) = \frac{2}{3}\frac{\overline{K}}{V}\sum_i N_i = \frac{2}{3}\left(\frac{N}{V}\right)\left(\overline{K}\right)$$

where N is the total number of molecules of all types and we have used the fact that the average kinetic energies of all types of molecules are the same because all the gases have the same temperature. The final expression for the pressure is the same as that of a single gas with N molecules in the same volume V and at the given temperature.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 21.1 Molecular Model of an Ideal Gas

P21.1 (a) The volume is $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(0.150\text{ m})^3 = 1.41\times10^{-2}\text{ m}^3$. The quantity of gas can be obtained from $PV = nRT$:

$$n = \frac{PV}{RT} = \frac{(1.013\times10^5\text{ N/m}^2)(1.41\times10^{-2}\text{ m}^3)}{(8.314\text{ N}\cdot\text{m/mol}\cdot\text{K})(293\text{ K})} = 0.588\text{ mol}$$

The number of molecules is

$$N = nN_A = (0.588\text{ mol})(6.02\times10^{23}\text{ molecules/mol})$$

$$N = \boxed{3.54\times10^{23}\text{ helium atoms}}$$

(b) The kinetic energy is given by $\bar{K} = \frac{1}{2}m_0\overline{v^2} = \frac{3}{2}k_BT$:

$$\bar{K} = \frac{3}{2}(1.38\times10^{-23}\text{ J/K})(293\text{ K}) = \boxed{6.07\times10^{-21}\text{ J}}$$

(c) An atom of He has mass

$$m_0 = \frac{M}{N_A} = \frac{4.002\ 6\text{ g/mol}}{6.02\times10^{23}\text{ molecules/mol}}$$
$$= 6.65\times10^{-24}\text{ g} = 6.65\times10^{-27}\text{ kg}$$

So the root-mean-square speed is given by

$$v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{2\bar{K}}{m_0}} = \sqrt{\frac{2\times6.07\times10^{-27}\text{ J}}{6.65\times10^{-27}\text{ kg}}} = \boxed{1.35\text{ km/s}}$$

P21.2 (a) Both kinds of molecules have the same average kinetic energy. It is

$$\bar{K} = \frac{3}{2}k_BT = \frac{3}{2}(1.38\times10^{-23}\text{ J/K})(423\text{ K}) = \boxed{8.76\times10^{-21}\text{ J}}$$

(b) The root-mean square velocity can be calculated from the kinetic energy:

$$v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{2\bar{K}}{m_0}}$$

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so $v_{\text{rms}} = \sqrt{\dfrac{1.75 \times 10^{-20}\ \text{J}}{m_0}}$ **[1]**

For helium,

$$m_0 = \frac{4.00\ \text{g/mol}}{6.02 \times 10^{23}\ \text{molecules/mol}} = 6.64 \times 10^{-24}\ \text{g/molecule}$$

$$m_0 = 6.64 \times 10^{-27}\ \text{kg/molecule}$$

Similarly for argon, $m_0 = \dfrac{39.9\ \text{g/mol}}{6.02 \times 10^{23}\ \text{molecules/mol}}$

$$= 6.63 \times 10^{-23}\ \text{g/molecule}$$

$$m_0 = 6.63 \times 10^{-26}\ \text{kg/molecule}$$

Substituting into [1] above,

we find for helium, $\boxed{v_{\text{rms}} = 1.62\ \text{km/s}}$

and for argon, $\boxed{v_{\text{rms}} = 514\ \text{m/s}}$

P21.3 (a) From Newton's second law, the average force is given by

$$\bar{F} = Nm\frac{\Delta v}{\Delta t} = 500\left(5.00 \times 10^{-3}\ \text{kg}\right) \times \frac{\left[8.00 \sin 45.0° - \left(-8.00 \sin 45.0°\right)\right]\ \text{m/s}}{30.0\ \text{s}}$$

$$= \boxed{0.943\ \text{N}}$$

(b) We find the pressure from

$$P = \frac{\bar{F}}{A} = \frac{0.943\ \text{N}}{0.600\ \text{m}^2} = 1.57\ \text{N/m}^2 = \boxed{1.57\ \text{Pa}}$$

P21.4 The equation of state for an ideal gas can be used with the given information to find the number of molecules in a specific volume.

$$PV = \left(\frac{N}{N_A}\right)RT \text{ means } N = \frac{PVN_A}{RT},$$

so that, suppressing units,

$$N = \frac{\left(1.00 \times 10^{-10}\right)(133)(1.00)\left(6.02 \times 10^{23}\right)}{(8.314)(300)}$$

$$= \boxed{3.21 \times 10^{12}\ \text{molecules}}$$

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P21.5 The gas temperature must be that implied by $\frac{1}{2}m_0\overline{v^2} = \frac{3}{2}k_BT$ for a monatomic gas like helium.

$$T = \frac{2}{3}\left(\frac{\frac{1}{2}m_0\overline{v^2}}{k_B}\right) = \frac{2}{3}\left(\frac{3.60\times10^{-22}\text{ J}}{1.38\times10^{-23}\text{ J/K}}\right) = 17.4\text{ K}$$

Now $PV = nRT$ gives

$$n = \frac{PV}{RT} = \frac{\left(1.20\times10^5\text{ N/m}^2\right)\left(4.00\times10^{-3}\text{ m}^3\right)}{(8.314\text{ J/mol}\cdot\text{K})(17.4\text{ K})} = \boxed{3.32\text{ mol}}$$

P21.6 $P = \frac{2}{3}\frac{N}{V}\overline{K}$ from the kinetic-theory account for pressure.

$$N = \frac{3}{2}\frac{PV}{\overline{K}}$$

$$n = \frac{N}{N_A} = \boxed{\frac{3}{2}\frac{PV}{\overline{K}N_A}}$$

P21.7 Use the equation describing the kinetic-theory account for pressure: $P = \frac{2N}{3V}\left(\frac{m_0\overline{v^2}}{2}\right)$. Then

$$\overline{K} = \frac{m_0\overline{v^2}}{2} = \frac{3PV}{2N}, \text{ where } N = nN_A$$

$$\overline{K} = \frac{3PV}{2nN_A} = \frac{3(8.00\text{ atm})\left(1.013\times10^5\text{ Pa/atm}\right)\left(5.00\times10^{-3}\text{ m}^3\right)}{2(2\text{ mol})\left(6.02\times10^{23}\text{ molecules/mol}\right)}$$

$$\overline{K} = \boxed{5.05\times10^{-21}\text{ J}}$$

P21.8 The molar mass of diatomic oxygen is 32.0 g. The rms speed of oxygen molecules is

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

and

$$p_{\text{rms}} = mv_{\text{rms}} = \frac{M}{N_A}\sqrt{\frac{3RT}{M}} = \frac{1}{N_A}\sqrt{3RTM}$$

$$= \frac{1}{6.02\times10^{23}}\sqrt{3\left(8.314\text{ J/mol}\cdot\text{K}\right)\left(350\text{ K}\right)\left(32.0\times10^{-3}\text{ kg}\right)}$$

$$p_{\text{rms}} = \boxed{2.78\times10^{-23}\text{ kg}\cdot\text{m/s}}$$

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P21.9 We use $1\text{ u} = 1.66 \times 10^{-24}\text{ g}$.

(a) For He, $m_0 = 4.00\text{ u}\left(\dfrac{1.66\times10^{-24}\text{ g}}{1\text{ u}}\right) = \boxed{6.64\times10^{-27}\text{ kg}}$

(b) For Fe, $m_0 = 55.9\text{ u}\left(\dfrac{1.66\times10^{-24}\text{ g}}{1\text{ u}}\right) = \boxed{9.28\times10^{-26}\text{ kg}}$

(c) For Pb, $m_0 = 207\text{ u}\left(\dfrac{1.66\times10^{-24}\text{ g}}{1\text{ u}}\right) = \boxed{3.44\times10^{-25}\text{ kg}}$

P21.10 The rms speed of molecules in a gas of molecular weight M and absolute temperature T is $v_{\text{rms}} = \sqrt{3RT/M}$. Thus, if $v_{\text{rms}} = 625$ m/s for molecules in oxygen (O_2), for which $M = 32.0$ g/mol = 32.0×10^{-3} kg/mol, the temperature of the gas is

$$T = \frac{Mv_{\text{rms}}^2}{3R} = \frac{\left(32.0\times10^{-3}\text{ kg/mol}\right)\left(625\text{ m/s}\right)^2}{3\left(8.31\text{ J/mol}\cdot\text{K}\right)} = \boxed{501\text{ K}}$$

***P21.11** (a) From the ideal gas law,

$$PV = nRT = \frac{Nm_0v^2}{3}$$

The total translational kinetic energy is $\dfrac{Nm_0v^2}{2} = E_{\text{trans}}$:

$$E_{\text{trans}} = \frac{3}{2}PV = \frac{3}{2}\left(3.00\times1.013\times10^5\text{ Pa}\right)\left(5.00\times10^{-3}\text{ m}^3\right)$$
$$= \boxed{2.28\text{ kJ}}$$

(b) $$\frac{m_0v^2}{2} = \frac{3k_BT}{2} = \frac{3RT}{2N_{\text{A}}} = \frac{3(8.314\text{ J/mol}\cdot\text{K})(300\text{ K})}{2\left(6.02\times10^{23}\right)} = \boxed{6.21\times10^{-21}\text{ J}}$$

P21.12 (a) The volume occupied by this gas is

$$V = 7.00\text{ L}\left(10^3\text{ cm}^3/1\text{ L}\right)\left(1\text{ m}^3/10^6\text{ cm}^3\right) = 7.00\times10^{-3}\text{ m}^3$$

Then, the ideal gas law gives

$$T = \frac{PV}{nR} = \frac{\left(1.60\times10^6\text{ Pa}\right)\left(7.00\times10^{-3}\text{ m}^3\right)}{\left(3.50\text{ mol}\right)\left(8.31\text{ J/mol}\cdot\text{K}\right)} = \boxed{385\text{ K}}$$

(b) The average kinetic energy per molecule in this gas is

$$\overline{KE}_{\text{molecule}} = \frac{3}{2}k_BT = \frac{3}{2}\left(1.38\times10^{-23}\text{ J/K}\right)\left(385\text{ K}\right) = \boxed{7.97\times10^{-21}\text{ J}}$$

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(c) You would need to know the mass of the gas molecule to find its average speed, which in turn requires knowledge of
the molecular mass of the gas.

P21.13 To find the pressure exerted by the nitrogen molecules, we first calculated the average force exerted by the molecules:

$$\bar{F} = Nm_0\frac{\Delta v}{\Delta t} = \frac{\left(5.00\times10^{23}\right)\left[\left(4.65\times10^{-26}\text{ kg}\right)2\left(300\text{ m/s}\right)\right]}{1.00\text{ s}} = 14.0\text{ N}$$

the pressure is then

$$P = \frac{\bar{F}}{A} = \frac{14.0\text{ N}}{8.00\times10^{-4}\text{ m}^2} = \boxed{17.4\text{ kPa}}$$

Section 21.2 Molar Specific Heat of an Ideal Gas

P21.14 $n = 1.00$ mol, $T_i = 300$ K

(a) Since V = constant, $W = \boxed{0}$.

(b) $\Delta E_{\text{int}} = Q + W = 209\text{ J} + 0 = \boxed{209\text{ J}}$

(c) $\Delta E_{\text{int}} = nC_V\Delta T = n\left(\frac{3}{2}R\right)\Delta T$

$$\text{so}\quad \Delta T = \frac{2\Delta E_{\text{int}}}{3nR} = \frac{2(209\text{ J})}{3(1.00\text{ mol})\left(8.314\text{ J/mol}\cdot\text{K}\right)} = 16.8\text{ K}$$

$$T = T_i + \Delta T = 300\text{ K} + 16.8\text{ K} = \boxed{317\text{ K}}$$

P21.15 $Q = \left(nC_p\Delta T\right)_{\text{isobaric}} + \left(nC_V\Delta T\right)_{\text{isovolumetric}}$

In the isobaric process, V doubles so T must double, to $2T_i$.

In the isovolumetric process, P triples so T changes from $2T_i$ to $6T_i$.

$$Q = n\left(\frac{7}{2}R\right)\left(2T_i - T_i\right) + n\left(\frac{5}{2}R\right)\left(6T_i - 2T_i\right) = 13.5nRT_i$$

$$= \boxed{13.5PV}$$

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P21.16 (a) Consider warming it at constant pressure. Oxygen and nitrogen are diatomic, so $C_P = \frac{7R}{2}$. Then,

$$Q = nC_P\Delta T = \frac{7}{2}nR\Delta T = \frac{7}{2}\left(\frac{PV}{T}\right)\Delta T$$

$$Q = \frac{7}{2}\frac{\left(1.013\times10^5\ \text{N/m}^2\right)\left(100\ \text{m}^3\right)}{300\ \text{K}}(1.00\ \text{K}) = \boxed{118\ \text{kJ}}$$

(b) We use the definition of gravitational potential energy,

$$U_g = mgy$$

from which,

$$m = \frac{U_g}{gy} = \frac{1.18\times10^5\ \text{J}}{\left(9.80\ \text{m/s}^2\right)(2.00\ \text{m})} = \boxed{6.03\times10^3\ \text{kg}}$$

P21.17 We use the tabulated values for C_P and C_V:

(a) Since this is a constant-pressure process, $Q = nC_P\Delta T$.

The temperature rises by ΔT = 420 K – 300 K = 120 K:

$$Q = nC_P\Delta T = (1.00\ \text{mol})\left(28.8\ \text{J/mol}\cdot\text{K}\right)(420\ \text{K} - 300\ \text{K})$$

$$= \boxed{3.46\ \text{kJ}}$$

(b) For any gas $\Delta E_{\text{int}} = nC_V\Delta T$, so

$$\Delta E_{\text{int}} = nC_V\Delta T = (1.00\ \text{mol})\left(20.4\ \text{J/mol}\cdot\text{K}\right)(120\ \text{K}) = \boxed{2.45\ \text{kJ}}$$

(c) The first law says $\Delta E_{\text{int}} = Q + W$, so

$$W = -Q + \Delta E_{\text{int}} = -3.46\ \text{kJ} + 2.45\ \text{kJ} = \boxed{-1.01\ \text{kJ}}$$

P21.18 (a) Molar specific heat is $C_V = \frac{5}{2}R$.

Specific heat at constant volume per unit mass is given by

$$c_V = \frac{C_V}{M} = \frac{5}{2}R\left(\frac{1}{M}\right)$$

$$= \frac{5}{2}\left(8.314\ \text{J/mol}\cdot\text{K}\right)\left(\frac{1.00\ \text{mol}}{0.028\ 9\ \text{kg}}\right)$$

$$= 719\ \text{J/kg}\cdot\text{K} = \boxed{0.719\ \text{kJ/kg}\cdot\text{K}}$$

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(b) $m = Mn = M\left(\frac{PV}{RT}\right)$

$$m = (0.028\,9\ \text{kg/mol})\left[\frac{(200\times10^3\ \text{Pa})(0.350\ \text{m}^3)}{(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K})}\right] = \boxed{0.811\ \text{kg}}$$

(c) We consider a constant-volume process where no work is done.

$$Q = mc_V\Delta T$$
$$= (0.811\ \text{kg})(0.719\ \text{kJ/kg}\cdot\text{K})(700\ \text{K} - 300\ \text{K})$$
$$= \boxed{233\ \text{kJ}}$$

(d) We now consider a constant-pressure process where the internal energy of the gas is increased and work is done.

$$Q = nC_P\Delta T = \frac{m}{M}(C_V + R)\Delta T = \frac{m}{M}\left(\frac{5}{2}R + R\right)\Delta T = \frac{m}{M}\left(\frac{7}{2}R\right)\Delta T$$

$$= m\left(\frac{7}{5}\right)\left(\frac{\frac{5}{2}R}{M}\right)\Delta T = m\left[\left(\frac{7}{5}\right)\left(\frac{C_V}{M}\right)\right]\Delta T$$

$$\rightarrow Q = (0.811\ \text{kg})\left[\frac{7}{5}(0.719\ \text{kJ/kg}\cdot\text{K})\right](400\ \text{K}) = \boxed{327\ \text{kJ}}$$

***P21.19** $\Delta E_{\text{int}} = \frac{3}{2}nR\Delta T = \frac{3}{2}(3.00\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(2.00\ \text{K}) = \boxed{74.8\ \text{J}}$

P21.20 Consider 800 cm^3 of tea (flavored water) at 90.0°C mixing with 200 cm^3 of diatomic ideal gas at 20.0°C:

$$Q_{\text{cold}} = -Q_{\text{hot}}$$

or $$m_{\text{air}}c_{P,\,\text{air}}\left(T_f - T_{i,\,\text{air}}\right) = -m_w c_w (\Delta T)_w$$

$$(\Delta T)_w = \frac{-m_{\text{air}}c_{P,\,\text{air}}\left(T_f - T_{i,\,\text{air}}\right)}{m_w c_w} = \frac{-(\rho V)_{\text{air}}\,c_{P,\,\text{air}}(90.0°\text{C} - 20.0°\text{C})}{(\rho_w V_w)c_w}$$

where we have anticipated that the final temperature of the mixture will be close to 90.0°C.

The molar specific heat of air is $C_{P,\,\text{air}} = \frac{7}{2}R$.

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So the specific heat per gram is

$$c_{P,\,\text{air}} = \frac{7}{2}\left(\frac{R}{M}\right) = \frac{7}{2}\left(8.314\ \text{J/mol}\cdot\text{K}\right)\left(\frac{1.00\ \text{mol}}{28.9\ \text{g}}\right) = 1.01\ \text{J/g}\cdot\text{°C}$$

and

$$(\Delta T)_w = -\frac{\left[\left(1.20\times10^{-3}\ \text{g/cm}^3\right)\left(200\ \text{cm}^3\right)\right](1.01\ \text{J/g}\cdot\text{°C})(70.0\text{°C})}{\left[\left(1.00\ \text{g/cm}^3\right)\left(800\ \text{cm}^3\right)\right](4.186\ \text{J/g}\cdot\text{°C})}$$

or $$(\Delta T)_w \approx -5.05\times10^{-3}\text{°C}$$

The change of temperature for the water is

between 10^{-3} °C and 10^{-2} °C.

***P21.21** (a) The air is far from liquefaction so it behaves as an ideal gas. From $PV = nRT$ we have $PV = \frac{m}{M}RT$, or $PM = \frac{m}{V}RT = \rho RT$. For the samples of air in the balloon at 10.0°C (cold) and at the elevated temperature (hot) we have $PM = \rho_c RT_c$ and $PM = \rho_h RT_h$. Then $\rho_h T_h = \rho_c T_c$ and $\rho_h = \frac{\rho_c T_c}{T_h}$. For equilibrium of the balloon on the point of rising,

$$\sum F_y = ma_y: \qquad +B - F_{g\ \text{hot air}} - F_{g\ \text{cargo}} = 0$$

$$+\rho_c Vg - \rho_h Vg - mg = 0$$

$$+\rho_c V - \frac{\rho_c T_c}{T_h}V - m = 0$$

$$\left(1.25\ \text{kg/m}^3\right)\left(400\ \text{m}^3\right) - \left(1.25\ \text{kg/m}^3\right)\left(\frac{283\ \text{K}}{T_h}\right)\left(400\ \text{m}^3\right) - 200\ \text{kg} = 0$$

$$300\ \text{kg} = (500\ \text{kg})\left(\frac{283\ \text{K}}{T_h}\right)$$

$$T_h = \left(\frac{500}{300}\right)(283\ \text{K}) = 472\ \text{K}$$

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The quantity of air that must be warmed is given by $PV = n_h RT_h$, or $n_h = \dfrac{PV}{RT_h}$. The heat input required is

$$\begin{aligned} Q &= nC_p \Delta T \\ &= -\frac{PV}{RT_h}\ \frac{7}{2} R(T_h - T_c) \\ &= \left(\frac{7}{2}\right)\frac{(1.013\times 10^5\ \text{N/m}^2)(400\ \text{m}^3)(472\ \text{K} - 283\ \text{K})}{472\ \text{K}} \\ &= \boxed{5.66\times 10^7\ \text{J}} \end{aligned}$$

(b) $Q = mH$, so $m = \dfrac{Q}{H} = \dfrac{5.66\times 10^7\ \text{J}}{5.03\times 10^7\ \text{J/kg}} = \boxed{1.12\ \text{kg}}$

Section 21.3 The Equipartition of Energy

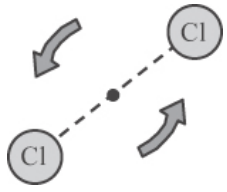
P21.22 (a) $E_{\text{int}} = Nf\left(\dfrac{k_B T}{2}\right) = f\left(\dfrac{nRT}{2}\right)$

(b) $C_V = \dfrac{1}{n}\left(\dfrac{dE_{\text{int}}}{dT}\right) = \dfrac{1}{2} fR$

(c) $C_P = C_V + R = \dfrac{1}{2}(f+2)R$

(d) $\gamma = \dfrac{C_P}{C_V} = \dfrac{f+2}{f}$

P21.23 The rotational kinetic energy of the molecule is given by $K_{\text{rot}} = \dfrac{1}{2} I\omega^2$. We determine the moment of inertia from $I = 2m_0 r^2$, with $m_0 = 35.0\times 1.67\times 10^{-27}$ kg and $r = 10^{-10}$ m:



ANS. FIG. P21.23

$$\begin{aligned} I &= 2m_0 r^2 = 2(35.0\times 1.67\times 10^{-27}\ \text{kg})(10^{-10}\ \text{m})^2 \\ &= 1.17\times 10^{-45}\ \text{kg}\cdot\text{m}^2 \end{aligned}$$

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Then,

$$K_{\text{rot}} = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(1.17\times10^{-45}\ \text{kg}\cdot\text{m}^2\right)\left(2.00\times10^{12}\ \text{s}^{-1}\right)^2$$

$$= \boxed{2.34\times10^{-21}\ \text{J}}$$

P21.24 We must have the difference of molar specific heats given by Equation 21.31: $C_P - C_V = R$. The value of γ tells us that $C_P = 1.75C_V$, so

$$1.75C_V - C_V = R \quad \rightarrow \quad C_V = \frac{R}{0.75} = \frac{4}{3}R$$

The maximum possible value of $\gamma = 1 + \frac{R}{C_V} = 1.67$ occurs for the lowest possible value for $C_V = \frac{3}{2}R$. Therefore the claim of $\gamma = 1.75$ for the newly discovered gas cannot be true.

P21.25 The sample's total heat capacity at constant volume is nC_v. An ideal gas of diatomic molecules has three degrees of freedom for translation in the x, y, and z directions. If we take the y axis along the axis of a molecule, then outside forces cannot excite rotation about this axis, since they have no lever arms. Collisions will set the molecule spinning only about the x and z axes.

(a) If the molecules do not vibrate, they have five degrees of freedom. Random collisions put equal amounts of energy $\frac{1}{2}k_BT$ into all five kinds of motion. The average energy of one molecule is $\frac{5}{2}k_BT$. The internal energy of the two-mole sample is

$$N\left(\frac{5}{2}k_BT\right) = nN_A\left(\frac{5}{2}k_BT\right) = n\left(\frac{5}{2}R\right)T = nC_VT$$

The molar heat capacity is $C_V = \frac{5}{2}R$, and the sample's heat capacity is

$$nC_V = n\left(\frac{5}{2}R\right) = (2.00\ \text{mol})\left[\frac{5}{2}\left(8.314\ \text{J/mol}\cdot\text{K}\right)\right]$$

$$\boxed{nC_V = 41.6\ \text{J/K}}$$

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(b) For the heat capacity at constant pressure, we have

$$nC_P = n\left(C_V + R\right) = n\left(\frac{5}{2}R + R\right) = \frac{7}{2}nR$$

$$= \left(2.00\text{ mol}\right)\left[\frac{7}{2}\left(8.314\text{ J/mol}\cdot\text{K}\right)\right]$$

$$\boxed{nC_P = 58.2\text{ J/K}}$$

(c) Vibration adds two more degrees of freedom for two more terms in the molecular energy, for kinetic and for elastic potential energy. We have

$$nC_V = n\left(\frac{7}{2}R\right) = \boxed{58.2\text{ J/K}}$$

and $nC_P = n\left(\frac{9}{2}R\right) = \boxed{74.8\text{ J/K}}$.

Section 21.4 Adiabatic Processes for an Ideal Gas

P21.26 (a) In an adiabatic process $P_iV_i^\gamma = P_fV_f^\gamma$:

$$P_f = P_i\left(\frac{V_i}{V_f}\right)^\gamma = \left(5.00\text{ atm}\right)\left(\frac{12.0}{30.0}\right)^{1.40} = \boxed{1.39\text{ atm}}$$

(b) The initial temperature is

$$T_i = \frac{P_iV_i}{nR} = \frac{5.00\left(1.013\times10^5\text{ Pa}\right)\left(12.0\times10^{-3}\text{ m}^3\right)}{\left(2.00\text{ mol}\right)\left(8.314\text{ J/mol}\cdot\text{K}\right)} = \boxed{366\text{ K}}$$

and similarly the final temperature is

$$T_f = \frac{P_fV_f}{nR} = \frac{1.39\left(1.013\times10^5\text{ Pa}\right)\left(30.0\times10^{-3}\text{ m}^3\right)}{\left(2.00\text{ mol}\right)\left(8.314\text{ J/mol}\cdot\text{K}\right)} = \boxed{253\text{ K}}$$

(c) The process is adiabatic, so by definition, $\boxed{Q = 0}$.

(d) For any process, $\Delta E_{\text{int}} = nC_V\Delta T$,

and for this diatomic ideal gas, $C_V = \frac{R}{\gamma - 1} = \frac{5}{2}R$

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Thus,

$$\Delta E_{\text{int}} = nC_V\Delta T$$
$$= (2.00\text{ mol})\left[\frac{5}{2}(8.314\text{ J/mol}\cdot\text{K})\right](253\text{ K} - 366\text{ K})$$
$$= \boxed{-4.66\text{ kJ}}$$

(e) $W = \Delta E_{\text{int}} - Q = -4.66\text{ kJ} - 0 = \boxed{-4.66\text{ kJ}}$

P21.27 (a) $P_iV_i^\gamma = P_fV_f^\gamma$ so $\frac{V_f}{V_i} = \left(\frac{P_i}{P_f}\right)^{1/\gamma} = \left(\frac{1.00}{20.0}\right)^{5/7} = \boxed{0.118}$.

(b) $\frac{T_f}{T_i} = \frac{P_fV_f}{P_iV_i} = \left(\frac{P_f}{P_i}\right)\left(\frac{V_f}{V_i}\right) = (20.0)(0.118) \rightarrow \frac{T_f}{T_i} = \boxed{2.35}$

(c) Since the process is adiabatic, $\boxed{Q = 0}$.

(d) Since $\gamma = 1.40 = \frac{C_P}{C_V} = \frac{R + C_V}{C_V}$, $C_V = \frac{5}{2}R$

and $\Delta T = 2.35T_i - T_i = 1.35T_i$, then

$$\Delta E_{\text{int}} = nC_V\Delta T$$
$$= (0.016\,0\text{ mol})\left(\frac{5}{2}\right)(8.314\text{ J/mol}\cdot\text{K})[1.35(300\text{ K})]$$
$$= \boxed{135\text{ J}}$$

(e) $W = -Q + \Delta E_{\text{int}} = 0 + 135\text{ J} = \boxed{+135\text{ J}}$

P21.28 (a) The work done *on* the gas is

$$W_{ab} = -\int_{V_a}^{V_b} P\,dV$$

For the isothermal process,

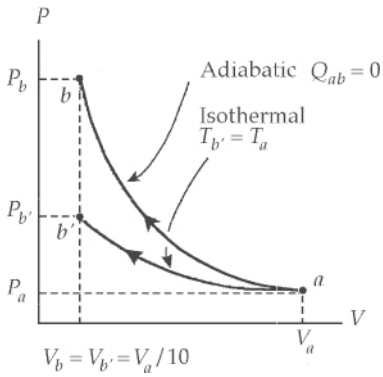
$$W_{ab'} = -nRT_a\int_{V_a}^{V_{b'}}\left(\frac{1}{V}\right)dV$$

$$W_{ab'} = -nRT_a\ln\left(\frac{V_{b'}}{V_a}\right) = nRT\ln\left(\frac{V_a}{V_{b'}}\right)$$

Thus,

$$W_{ab'} = (5.00\text{ mol})(8.314\text{ J/mol}\cdot\text{K})(293\text{ K})\ln(10.0) = \boxed{28.0\text{ kJ}}$$

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ANS. FIG. P21.28

(b) For the adiabatic process, we must first find the final temperature, T_b. Since air consists primarily of diatomic molecules, we shall use

$$\gamma_{\text{air}} = 1.40 \quad \text{and} \quad C_{V,\text{air}} = \frac{5R}{2} = \frac{5(8.314)}{2} = 20.8 \text{ J/mol} \cdot \text{K}$$

Then, for the adiabatic process,

$$T_b = T_a \left(\frac{V_a}{V_b} \right)^{\gamma - 1} = (293 \text{ K})(10.0)^{0.400} = 736 \text{ K}$$

Thus, the work done on the gas during the adiabatic process is

$$W_{ab} \left(-Q + \Delta E_{\text{int}} \right)_{ab} = \left(-0 + nC_V \Delta T \right)_{ab} = nC_V \left(T_b - T_a \right)$$

or $$W_{ab} = (5.00 \text{ mol})(20.8 \text{ J/mol} \cdot \text{K})(736 \text{ K} - 293 \text{ K}) = \boxed{46.0 \text{ kJ}}$$

(c) For the isothermal process, we have $P_{b'} V_{b'} = P_a V_a$.

Thus, $P_{b'} = P_a \left(\frac{V_a}{V_{b'}} \right) = (1.00 \text{ atm})(10.0) = \boxed{10.0 \text{ atm}}$.

(d) For the adiabatic process, we have $P_b V_b^{\gamma} = P_a V_a^{\gamma}$.

Thus, $P_b = P_a \left(\frac{V_a}{V_b} \right)^{\gamma} = (1.00 \text{ atm})(10.0)^{1.40} = \boxed{25.1 \text{ atm}}$.

P21.29 Combining PV^{γ} = constant with the ideal gas law gives one of the textbook equations describing adiabatic processes, $T_1 V_1^{\gamma - 1} = T_2 V_2^{\gamma - 1}$.

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma - 1} = (300 \text{ K}) \left(\frac{1}{2} \right)^{(1.40 - 1)} = \boxed{227 \text{ K}}$$

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P21.30 Use Equation 21.37 for an adiabatic process to find the temperature of the compressed fuel-air mixture at the end of the compression stroke, before ignition:

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}$$

which gives

$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{\gamma-1} = (323 \text{ K})(14.5)^{1.40-1} = 941 \text{ K}$$

This is equivalent to 668°C, which is higher than the melting point of aluminum which is 660°C. Also, the temperature will rise much more when ignition occurs. The engine will melt when put into operation!Therefore, the claim of improved efficiency using an engine fabricated out of aluminum cannot be true.

P21.31 We suppose the air plus burnt gasoline behaves like a diatomic ideal gas. We find its final absolute pressure:

$$(21.0 \text{ atm})\left(50.0 \text{ cm}^3\right)^{7/5} = P_f \left(400 \text{ cm}^3\right)^{7/5}$$

$$P_f = (21.0 \text{ atm})\left(\frac{1}{8}\right)^{7/5} = 1.14 \text{ atm}$$

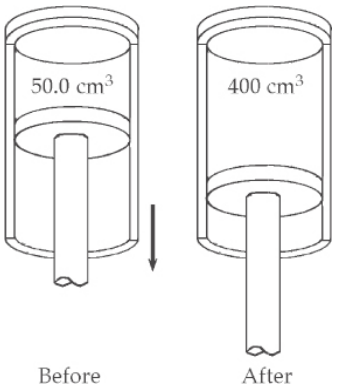
Now $Q = 0$ and $W = \Delta E_{\text{int}} = nC_V \left(T_f - T_i\right)$,

so $$W = \frac{5}{2} nRT_f - \frac{5}{2} nRT_i = \frac{5}{2}\left(P_f V_f - P_i V_i\right)$$

$$W = \frac{5}{2}[(1.14 \text{ atm})(400 \text{ cm}^3) - (21.0 \text{ atm})(50.0 \text{ cm}^3)]$$

$$= \left(-1\ 485 \text{ atm} \cdot \text{cm}^3\right)\left(\frac{1.013 \times 10^5 \text{ N/m}^2}{1 \text{ atm}} \right)\left(10^{-6} \text{ m}^3/\text{cm}^3\right)$$

$$= -150 \text{ J}$$



ANS. FIG. P21.31

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The output work is $-W = +150\ \text{J}$

The time for this stroke is $\frac{1}{4}\left(\frac{1\ \text{min}}{2\,500}\right)\left(\frac{60\ \text{s}}{1\ \text{min}}\right) = 6.00\times10^{-3}\ \text{s}$

$$P = \frac{-W}{\Delta t} = \frac{150\ \text{J}}{6.00\times10^{-3}\ \text{s}} = \boxed{25.0\ \text{kW}}$$

P21.32 (a) $V_i = \pi\left(\frac{2.50\times10^{-2}\ \text{m}}{2}\right)^2(0.500\ \text{m}) = \boxed{2.45\times10^{-4}\ \text{m}^3}$

(b) The quantity of air we find from $P_iV_i = nRT_i$:

$$n = \frac{P_iV_i}{RT_i} = \frac{\left(1.013\times10^5\ \text{Pa}\right)\left(2.45\times10^{-4}\ \text{m}^3\right)}{\left(8.314\ \text{J/mol}\cdot\text{K}\right)(300\ \text{K})}$$

$$n = \boxed{9.97\times10^{-3}\ \text{mol}}$$

(c) Absolute pressure = gauge pressure + external pressure:

$$P_f = 101.3\ \text{kPa} + 800\ \text{kPa} = 901.3\ \text{kPa} = \boxed{9.01\times10^5\ \text{Pa}}$$

(d) Adiabatic compression: $P_iV_i^\gamma = P_fV_f^\gamma$

$$V_f = V_i\left(\frac{P_i}{P_f}\right)^{1/\gamma} = \left(2.45\times10^{-4}\ \text{m}^3\right)\left(\frac{101.3}{901.3}\right)^{5/7}$$

$$V_f = \boxed{5.15\times10^{-5}\ \text{m}^3}$$

(e) $P_fV_f = nRT_f$

$$T_f = T_i\frac{P_fV_f}{P_iV_i} = T_i\frac{P_f}{P_i}\left(\frac{P_i}{P_f}\right)^{1/\gamma} = T_i\left(\frac{P_i}{P_f}\right)^{(1/\gamma-1)}$$

$$T_f = 300\ \text{K}\left(\frac{101.3}{901.3}\right)^{(5/7-1)} = \boxed{560\ \text{K}}$$

(f) The work done on the gas in compressing it is $W = \Delta E_{\text{int}} = nC_V\Delta T$:

$$\Delta E_{\text{int}} = W = nC_V\Delta T$$

$$= \left(9.97\times10^{-3}\ \text{mol}\right)\frac{5}{2}\left(8.314\ \text{J/mol}\cdot\text{K}\right)(560\ \text{K} - 300\ \text{K})$$

$$\Delta E_{\text{int}} = \boxed{53.9\ \text{J}}$$

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(g) The pump wall has outer diameter 25.0 mm + 2.00 mm + 2.00 mm = 29.0 mm, and volume

$$\left[\pi\left(14.5\times10^{-3}\text{ m}\right)^2-\pi\left(12.5\times10^{-3}\text{ m}\right)^2\right]\times\left(4.00\times10^{-2}\text{ m}\right)=\boxed{6.79\times10^{-6}\text{ m}^3}$$

(h) The mass of the pump is given by

$$\rho V=\left(7.86\times10^{3}\text{ kg/m}^3\right)\left(6.79\times10^{-6}\text{ m}^3\right)=\boxed{53.3\text{ g}}$$

(i) Now imagine this energy being shared with the inner wall as the gas is held at constant volume. The overall warming process is described by

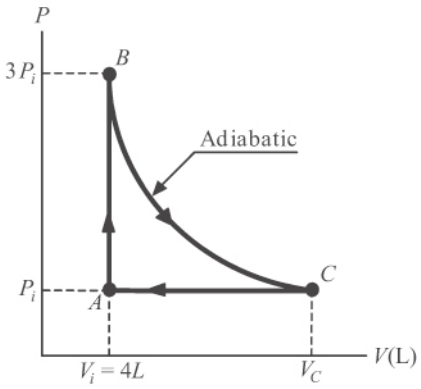
$$\Delta E_{\text{int}}=W=nC_V\Delta T+mc\Delta T\quad\rightarrow\quad\Delta T=\frac{W}{nC_V+mc}$$

Suppressing the units of R,

$$\Delta T=\frac{53.9\text{ J}}{\left(9.97\times10^{-3}\text{ mol}\right)\frac{5}{2}(8.314)+\left(0.053\,3\text{ kg}\right)\left(448\text{ J/kg}\cdot{}^\circ\text{C}\right)}$$

$$=2.24^\circ\text{C}=\boxed{2.24\text{ K}}$$

P21.33 (a) See ANS. FIG. P21.33(a) on the right.



ANS. FIG. P21.33(a)

(b) $P_BV_B^\gamma=P_CV_C^\gamma$

$3P_iV_i^\gamma=P_iV_C^\gamma$

$V_C=\left(3^{1/\gamma}\right)V_i=\left(3^{5/7}\right)V_i=2.19V_i$

$V_C=2.19(4.00\text{ L})=\boxed{8.77\text{ L}}$

(c) $P_BV_B=nRT_B=3P_iV_i=3nRT_i$

$T_B=3T_i=3(300\text{ K})=\boxed{900\text{ K}}$

(d) After one whole cycle, $T_A=T_i=\boxed{300\text{ K}}$.

(e) In AB, $Q_{AB}=nC_V\Delta V=n\left(\frac{5}{2}R\right)\left(3T_i-T_i\right)=(5.00)nRT_i$

$Q_{BC}=0$ as this process is adiabatic.

$P_CV_C=nRT_C=P_i\left(2.19V_i\right)=(2.19)nRT_i$

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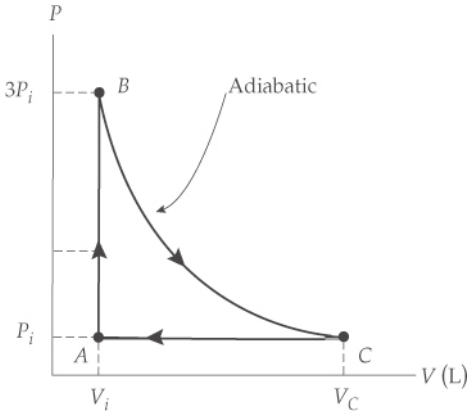
so $T_C = 2.19T_i$, and

$$Q_{CA} = nC_P\Delta T = n\left(\frac{7}{2}R\right)\left(T_i - 2.19T_i\right) = (-4.17)nRT_i$$

For the whole cycle,

$$Q_{ABCA} = Q_{AB} + Q_{BC} + Q_{CA} = (5.00 - 4.17)nRT_i = (0.829)nRT_i$$
$$\left(\Delta E_{\text{int}}\right)_{ABCA} = 0 = Q_{ABCA} + W_{ABCA}$$
$$W_{ABCA} = -Q_{ABCA} = -(0.829)nRT_i = -(0.829)P_iV_i$$
$$W_{ABCA} = -(0.829)\left(1.013\times10^5\ \text{Pa}\right)\left(4.00\times10^{-3}\ \text{m}^3\right) = \boxed{-336\ \text{J}}$$

P21.34 (a) Refer to ANS. FIG. P21.34(a).



ANS. FIG. P21.34(a)

(b) $P_BV_B^\gamma = P_CV_C^\gamma$

$3P_iV_i^\gamma = P_iV_C^\gamma$

$V_C = \boxed{\left(3^{1/\gamma}\right)V_i}$

(c) $P_BV_B = nRT_B = 3P_iV_i = 3nRT_i$

$T_B = \boxed{3T_i}$

(d) After one whole cycle,

$T_A = \boxed{T_i}$.

(e) For AB,

$$Q_{AB} = nC_v\Delta T = n\frac{R}{\gamma-1}\Delta T = n\frac{R}{\gamma-1}\left(3T_\text{i} - T_\text{i}\right) = \frac{2nRT_\text{i}}{\gamma-1} = \frac{2P_\text{i}V_\text{i}}{\gamma-1}$$

$Q_{BC} = 0$ as this process is abiabatic.

$$P_CV_C = nRT_C = P_i\left(3^{1/\gamma}\right)V_i = \left(3^{1/\gamma}\right)nRT_i \text{ so } T_C = \left(3^{1/\gamma}\right)T_i$$

$$Q_{CA} = nC_P\Delta T = n\gamma C_V\left[T_i - \left(3^{1/\gamma}\right)T_i\right] = \gamma\frac{R}{\gamma-1}nT_i\left[1-\left(3^{1/\gamma}\right)\right]$$
$$= P_iV_i\gamma\left(\frac{1}{\gamma-1}\right)\left[1-\left(3^{1/\gamma}\right)\right]$$

For the whole cycle,

$$Q_{ABCA} = Q_{AB} + Q_{BC} + Q_{CA} = \frac{2P_\text{i}V_\text{i}}{\gamma-1} + 0 + P_iV_i\gamma\left(\frac{1}{\gamma-1}\right)\left[1-3^{1/\gamma}\right]$$

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$$Q_{ABCA} = P_i V_i \left[\frac{2}{\gamma - 1} + \gamma \left(\frac{1}{\gamma - 1} \right) \left[1 - 3^{1/\gamma} \right] \right]$$

$$= P_i V_i \left[\frac{2}{\gamma - 1} + \left(\frac{\gamma - 1 + 1}{\gamma - 1} \right) \left[1 - 3^{1/\gamma} \right] \right]$$

$$Q_{ABCA} = P_i V_i \left[\frac{2}{\gamma - 1} + \left(1 - 3^{1/\gamma} \right) + \left(\frac{1 - 3^{1/\gamma}}{\gamma - 1} \right) \right]$$

$$= P_i V_i \left[\left(1 - 3^{1/\gamma} \right) + \left(\frac{3 - 3^{1/\gamma}}{\gamma - 1} \right) \right]$$

$$\left(\Delta E_{\text{int}} \right)_{ABCA} = 0 = Q_{ABCA} + W_{ABCA}$$

$$W_{ABCA} = -Q_{ABCA} = \boxed{-P_i V_i \left[\left(\frac{1}{\gamma - 1} \right) \left(1 - 3^{1/\gamma} \right) + \left(1 - 3^{1/\gamma} \right) \right]}$$

Section 21.5 Distribution of Molecular Speeds

***P21.35** The most probable speed is

$$v_{\text{mp}} = \sqrt{\frac{2k_B T}{m_0}} = \sqrt{\frac{2\left(1.38 \times 10^{-23}\ \text{J/K}\right)(4.20\ \text{K})}{6.64 \times 10^{-27}\ \text{kg}}} = \boxed{132\ \text{m/s}}$$

P21.36 (a) The average is

$$\bar{v} = \frac{\sum n_i v_i}{\sum n_i}$$

$$= \frac{1(2.00) + 2(3.00) + 3(5.00) + 4(7.00) + 3(9.00) + 2(12.0)}{1 + 2 + 3 + 4 + 3 + 2}\ \text{m/s}$$

$$\bar{v} = \boxed{6.80\ \text{m/s}}$$

(b) To find the average squared speed we work out

$$\overline{v^2} = \frac{\sum n_i v_i^2}{\sum n_i}$$

$$\overline{v^2} = \left(\frac{1}{15} \right) \left[1\left(2.00^2\right) + 2\left(3.00^2\right) + 3\left(5.00^2\right) + 4\left(7.00^2\right) + 3\left(9.00^2\right) + 2\left(12.0^2\right)\ \text{m}^2/\text{s}^2 \right]$$

$$\overline{v^2} = 54.9\ \text{m}^2/\text{s}^2$$

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Then the rms speed is

$$v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{54.9\ \text{m}^2/\text{s}^2} = \boxed{7.41\ \text{m/s}}$$

(c) More particles have $\boxed{v_{\text{mp}} = 7.00\ \text{m/s}}$ than any other speed.

P21.37 (a) The ratio of the number at higher energy to the number at lower energy is $e^{-\Delta E/k_B T}$, where ΔE is the energy difference. Here,

$$\Delta E = (10.2\ \text{eV})\left(\frac{1.60\times10^{-19}\ \text{J}}{1\ \text{eV}}\right) = 1.63\times10^{-18}\ \text{J}$$

and at 0°C,

$$k_B T = \left(1.38\times10^{-23}\ \text{J/K}\right)(273\ \text{K}) = 3.77\times10^{-21}\ \text{J}$$

Since this is much less than the excitation energy, nearly all the atoms will be in the ground state and the number excited is

$$\left(2.70\times10^{25}\right)\exp\left(\frac{-1.63\times10^{-18}\ \text{J}}{3.77\times10^{-21}\ \text{J}}\right) = \left(2.70\times10^{25}\right)e^{-433}$$

This number is much less than one, so
$\boxed{\text{almost all of the time no atom is excited.}}$

(b) At 10 000°C,

$$k_B T = \left(1.38\times10^{-23}\ \text{J/K}\right)(10\ 273\ \text{K}) = 1.42\times10^{-19}\ \text{J}$$

The number excited is

$$\left(2.70\times10^{25}\right)\exp\left(\frac{-1.63\times10^{-18}\ \text{J}}{1.42\times10^{-19}\ \text{J}}\right)$$

$$= \left(2.70\times10^{25}\right)e^{-11.5} = \boxed{2.70\times10^{20}}$$

P21.38 (a) $$\frac{V_{\text{rms, 35}}}{V_{\text{rms, 37}}} = \frac{\sqrt{3RT/M_{35}}}{\sqrt{3RT/M_{37}}} = \left(\frac{37.0\ \text{g/mol}}{35.0\ \text{g/mol}}\right)^{1/2} = \boxed{1.03}$$

(b) The lighter atom, $\boxed{^{35}\text{Cl}}$, moves faster.

P21.39 (a) From $v_{\text{avg}} = \sqrt{\dfrac{8k_B T}{\pi m_0}}$ we find the temperature as

$$T = \frac{\pi\left(6.64\times10^{-27}\ \text{kg}\right)\left(1.12\times10^{4}\ \text{m/s}\right)^2}{8\left(1.38\times10^{-23}\ \text{J/mol}\cdot\text{K}\right)} = \boxed{2.37\times10^{4}\ \text{K}}$$

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(b) $$T = \frac{\pi\left(6.64\times10^{-27}\ \text{kg}\right)\left(2.37\times10^{3}\ \text{m/s}\right)^2}{8\left(1.38\times10^{-23}\ \text{J/mol}\cdot\text{K}\right)} = \boxed{1.06\times10^{3}\ \text{K}}$$

P21.40 For a molecule of diatomic nitrogen the mass is

$$m_0 = \frac{M}{N_A} = \frac{28.0\times10^{-3}\ \text{kg/mol}}{6.02\times10^{23}\ \text{molecules/mol}}$$
$$= 4.65\times10^{-26}\ \text{kg/molecule}$$

(a) $$v_{\text{mp}} = \sqrt{\frac{2k_BT}{m_0}} = \sqrt{\frac{2(1.38\times10^{-23}\ \text{J/molecule}\cdot\text{K})(900\ \text{K})}{4.65\times10^{-26}\ \text{kg/molecule}}} = \boxed{731\ \text{m/s}}$$

(b) $$v_{\text{avg}} = \sqrt{\frac{8k_BT}{\pi m_0}} = \sqrt{\frac{8(1.38\times10^{-23}\ \text{J/molecule}\cdot\text{K})(900\ \text{K})}{\pi\cdot4.65\times10^{-26}\ \text{kg/molecule}}} = \boxed{825\ \text{m/s}}$$

(c) $$v_{\text{rms}} = \sqrt{\frac{3k_BT}{m_0}} = \sqrt{\frac{3(1.38\times10^{-23}\ \text{J/molecule}\cdot\text{K})(900\ \text{K})}{4.65\times10^{-26}\ \text{kg/molecule}}} = \boxed{895\ \text{m/s}}$$

(d) $\boxed{\text{The graph appears to be drawn correctly within about } 10\ \text{m/s}.}$

P21.41 (a) From the Boltzmann distribution law, the number density of molecules with height y so that the gravitational potential energy of the molecule-Earth system is m_0gy is $n_0e^{-m_0gy/k_BT}$. These are the molecules with height y, so this is the number per volume at height y as a function of y.

(b) $$\frac{n(y)}{n_0} = e^{-m_0gy/k_BT} = e^{-Mgy/N_Ak_BT} = e^{-Mgy/RT}$$

$$= e^{-\left(28.9\times10^{-3}\ \text{kg/mol}\right)\left(9.8\ \text{m/s}^2\right)\left(11\times10^{3}\ \text{m}\right)/\left(8.314\ \text{J/mol}\cdot\text{K}\right)(293\ \text{K})}$$
$$= e^{-1.279} = \boxed{0.278}$$

P21.42 In the Maxwell-Boltzmann speed distribution function take $\frac{dN_v}{dv} = 0$ to find

$$4\pi N\left(\frac{m_0}{2\pi k_BT}\right)^{3/2}\exp\left(-\frac{m_0v^2}{2k_BT}\right)\left(2v - \frac{2m_0v^3}{2k_BT}\right) = 0$$

and solve for v to find the most probable speed. Reject as solutions $v = 0$ and $v = \infty$. They describe minimally probable speeds.

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Retain only $2 - \frac{m_0 v^2}{k_B T} = 0.$

Then, $v_{\text{mp}} = \sqrt{\frac{2k_B T}{m_0}}.$

P21.43 It is convenient in the following to define $a = \frac{m_0 g}{k_B T}$.

(a) We calculate

$$\int_0^\infty e^{-m_0 g y / k_B T}\, dy = \int_0^\infty e^{-ay}\, dy = \int_{y=0}^\infty e^{-ay}\left(-a\, dy\right)\left(-\frac{1}{a}\right)$$

$$= \left(-\frac{1}{a}\right) e^{-ay}\Big|_0^\infty = \left(-\frac{1}{a}\right)(0-1) = \frac{1}{a}$$

Using Table B.6 in the appendix,

$$\int_0^\infty y e^{-ay} dy = \frac{1!}{(a)^2} = \left(\frac{1}{a}\right)^2$$

Then,

$$y_{\text{avg}} = \frac{\int_0^\infty y e^{-ay} dy}{\int_0^\infty e^{-ay} dy} = \frac{(1/a)^2}{1/a} = \frac{1}{a} = \frac{k_B T}{m_0 g}$$

(b) $$y_{\text{avg}} = \frac{k_B T}{(M/N_A) g} = \frac{RT}{Mg} = \frac{(8.314\ \text{J/mol}\cdot\text{K})(283\ \text{K})}{(28.9\times 10^{-3}\ \text{kg})(9.8\ \text{m/s}^2)} = \boxed{8.31\ \text{km}}$$

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Additional Problems

P21.44 (a) The average speed is given by $v_{\text{avg}} = \dfrac{\sum_i^N v_i}{N}$

which may be solved numerically for the values given. Suppressing units,

$$v_{\text{avg}} = \frac{\sum_i^N v_i}{N}$$

$$= \frac{\left[(3.00)+(4.00)+(5.80)+(2.50)+(3.60)+(1.90)+(3.80)+(6.60)\right]}{8}$$

$$= \frac{31.2 \text{ km/s}}{8} = \boxed{3.90 \text{ km/s}}$$

(b) The rms speed of the molecules is given by $v_{\text{rms}} = \sqrt{\dfrac{\sum_i^N v_i^2}{N}}$

which may be solved numerically for the values given. Suppressing units,

$$v_{\text{rms}} = \sqrt{\frac{\sum_i^N v_i^2}{N}}$$

$$= \sqrt{\frac{\left[(3.00)^2+(4.00)^2+(5.80)^2+(2.50)^2+(3.60)^2+(1.90)^2+(3.80)^2+(6.60)^2\right]}{8}}$$

$$= \sqrt{\frac{\left[(9.00)+(16.00)+(33.64)+(6.25)+(12.96)+(3.61)+(14.44)+(43.56)\right]}{8}}$$

$$= \sqrt{\frac{139.46 \text{ km}^2/\text{s}^2}{8}} = \sqrt{17.43 \text{ km}^2/\text{s}^2} = \boxed{4.18 \text{ km/s}}$$

P21.45 (a) The total amount of oxygen in the tank is (using $PV = nRT$)

$$n = \frac{PV}{RT} = \frac{(125 \text{ atm})(6.88 \text{ L})}{(0.0821 \text{ L}\cdot\text{atm/mol}\cdot\text{K})(21.0^\circ\text{C}+273)} = 35.6 \text{ mol}$$

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The rate at which the tank is being depleted (in moles/sec; again using $PV = nRT$) is

$$(\Delta n/\Delta t) = \frac{P(\Delta V/\Delta t)}{RT}$$
$$= \frac{(1\text{ atm})(8.50\text{ L/min})(1\text{min}/60\text{sec})}{(0.082\ 1\text{ L}\cdot\text{atm/mol}\cdot\text{K})(21.0°\text{C}+273)}$$
$$= 0.005\ 87\text{ mol/s}$$

The time interval to deplete the tank is the total molar mass divided by the molar rate:

$$\Delta t = n/(\Delta n/\Delta t) = \frac{35.6\text{ mol}}{5.87\times10^{-3}\text{ mol/s}}\left(\frac{1\text{ h}}{3\ 600\text{ s}}\right) = \boxed{1.70\text{ h}}$$

(b) Because the rms speed is dependent only on the molecular mass and the temperature, i.e.,

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

and because the masses of the molecule inside and at the outlet are the same, and the temperatures are the same (21.0°C), the rms speeds will be identical: the requested ratio is equal to $\boxed{1.00}$.

P21.46 (a) $$n = \frac{PV}{RT} = \frac{(1.013\times10^5\text{ Pa})(4.20\text{ m})(3.00\text{ m})(2.50\text{ m})}{(8.314\text{ J/mol}\cdot\text{K})(293\text{ K})} = 1.31\times10^3\text{ mol}$$

$$N = nN_A = \left(1.31\times10^3\text{ mol}\right)\left(6.02\times10^{23}\text{ molecules/mol}\right)$$

$$N = \boxed{7.89\times10^{26}\text{ molecules}}$$

(b) $$m = nM = \left(1.31\times10^3\text{ mol}\right)\left(0.028\ 9\text{ kg/mol}\right) = \boxed{37.9\text{ kg}}$$

(c) $$\frac{1}{2}m_0v^2 = \frac{3}{2}k_BT = \frac{3}{2}\left(1.38\times10^{-23}\text{ J/k}\right)(293\text{ K}) = \boxed{6.07\times10^{-21}\text{ J}}$$

(d) For one molecule,

$$m_0 = \frac{M}{N_A} = \frac{0.028\ 9\text{ kg/mol}}{6.02\times10^{23}\text{ molecules/mol}} = 4.80\times10^{-26}\text{ kg/molecule}$$

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{2\left(6.07\times10^{-21}\text{ J/molecule}\right)}{4.80\times10^{-26}\text{ kg/molecule}}} = \boxed{503\text{ m/s}}$$

(e) $\boxed{0}$

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(f) When the furnace operates, air expands and some of it leaves the room. The smaller mass of warmer air left in the room contains the same internal energy as the cooler air initially in the room.

P21.47 (a) The rms speed of molecules in a gas is related to the temperature by

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

which can be rearranged and solved numerically for the temperature:

$$T = \frac{mv_{\text{rms}}^2}{3k} = \frac{32\left(1.66\times10^{-27}\text{ kg}\right)\left(535\text{ m/s}\right)^2}{3\left(1.38\times10^{-23}\text{ J/K}\right)}$$

$$= \boxed{367\text{ K}}$$

(b) The rms speed is inversely related to the mass of the gas molecule (the mass is in the denominator of the square-root function above). The rms speed of nitrogen would be higher because the molar mass of nitrogen is less than that of oxygen.

(c) The rms speed of the nitrogen molecules is:

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3\left(1.38\times10^{-23}\text{ J/K}\right)(367\text{ K})}{28\left(1.66\times10^{-27}\text{ kg}\right)}} = \boxed{572\text{ m/s}}$$

P21.48 (a) The mean free path is given by:

$$\ell = \frac{1}{\sqrt{2}\pi d^2 N_V}$$

which can be solved numerically (noting that the pressure must be given as total pressure, not gauge pressure, and the temperature must be given in kelvins). Using $PV = nRT$:

$$N_V = N_A\left(\frac{n}{V}\right) = N_A\left(\frac{P}{RT}\right)$$

$$= \left(6.02\times10^{23}\text{ mol}^{-1}\right)\frac{(100\text{ atm}+1\text{ atm})\left(1.013\times10^5\ \frac{\text{N/m}^2}{\text{atm}}\right)}{(8.314\text{ J/mol}\cdot\text{K})(25.0^\circ\text{C}+273.15^\circ)}$$

$$= 2.48\times10^{27}\text{ molecules/m}^3$$

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which can then be inserted into the mean free path equation:

$$\ell = \frac{1}{\sqrt{2}\pi d^2 N_V} = \frac{1}{\sqrt{2}\pi\left(2.00\times10^{-10}\,\text{m}\right)^2\left(2.48\times10^{27}\,\text{mol/m}^3\right)}$$

$$= \boxed{2.26\times10^{-9}\ \text{m}}$$

(b) The average time interval between collisions is the inverse of the collision frequency:

$$t_{\text{collision}} = \frac{1}{f} = \frac{1}{\left(\frac{v_{\text{avg}}}{\ell}\right)} = \frac{\ell}{v_{\text{avg}}}$$

which can be solved numerically (where the value of v_{ave} is obtained from eqn. 21.26):

$$t_{\text{collision}} = \frac{\ell}{v_{\text{avg}}} = \frac{\ell}{\sqrt{\frac{8k_bT}{\pi m_0}}} = \frac{2.26\times10^{-9}\ \text{m}}{\sqrt{\frac{8\left(1.38\times10^{-23}\ \text{J/K}\right)(298.15\ \text{K})}{\pi\cdot32\left(1.66\times10^{-27}\ \text{kg}\right)}}}$$

$$= \frac{2.26\times10^{-9}\ \text{m}}{(444.1\ \text{m/s})} = \boxed{5.09\times10^{-12}\ \text{seconds}}$$

P21.49 For the system of the bullet, Equation 8.2 becomes

$$W_{\text{on bullet}} = \Delta K$$

For the system of the gas undergoing an adiabatic process, so that $Q = 0$, Equation 8.2 becomes

$$W_{\text{on gas}} = \Delta E_{\text{int}}$$

Recognizing that $W_{\text{on bullet}} = -W_{\text{on gas}}$, we see that

$$\Delta E_{\text{int}} = -\Delta K$$

Substituting for the internal energy, we find

$$nC_V\Delta T = -\Delta K$$

$$n\left(\frac{5}{2}R\right)\Delta T = -\Delta K \quad \rightarrow \quad \frac{5}{2}nR\left(T_f - T_i\right) = -\Delta K$$

Now use Equation 21.20 to substitute for T_f:

$$\frac{5}{2}nR\left[\left(\frac{V_i}{V_f}\right)^{\gamma-1}T_i - T_i\right] = -\Delta K$$

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$$\frac{5}{2}nRT_i\left[\left(\frac{V_i}{V_f}\right)^{\gamma-1}-1\right]=-\Delta K$$

$$\frac{5}{2}P_iV_i\left[\left(\frac{V_i}{V_f}\right)^{\gamma-1}-1\right]=-\left(\frac{1}{2}mv^2-0\right)$$

$$P_i=\frac{mv^2}{5V_i\left[1-\left(\frac{V_i}{V_f}\right)^{\gamma-1}\right]}$$

The final volume of the gas is equal to the initial volume plus the volume of the rifle barrel:

$$V_f=V_i+Ah=12.0\text{ cm}^3+\left(0.0300\text{ cm}^2\right)(50.0\text{ cm})=13.5\text{ cm}^3$$

Substituting numerical values,

$$P_i=\frac{\left(0.001\,10\text{ kg}\right)(120\text{ m/s})^2}{5\left(12\times10^{-6}\text{ m}^3\right)\left[1-\left(\frac{12.0\text{ cm}^3}{13.5\text{ cm}^3}\right)^{0.400}\right]}$$

$$=5.74\times10^6\text{ Pa}=\boxed{56.6\text{ atm}}$$

P21.50 (a) For a pure metallic element, one atom is one molecule. Its energy can be represented as

$$\frac{1}{2}m_0v_x^2+\frac{1}{2}m_0v_y^2+\frac{1}{2}m_0v_z^2+\frac{1}{2}k_xx^2+\frac{1}{2}k_yy^2+\frac{1}{2}k_zz^2$$

Its average value is

$$\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT=3k_BT$$

The energy of one mole is obtained by multiplying by Avogadro's number, $E_{\text{int}}/n = 3RT$, and the molar heat capacity at constant volume is $E_{\text{int}}/nT = 3R$.

(b) We calculate the specific heat from

$$3(8.314\text{ J/mol}\cdot\text{K})=\frac{3(8.314\text{ J})}{\left(55.845\times10^{-3}\text{ kg}\right)\cdot\text{K}}=\boxed{447\text{ J/kg}\cdot\text{K}}$$

This agrees with the tabulated value of 448 J/kg · °C within 0.3%.

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(c) For gold,

$$3(8.314\ \text{J/mol}\cdot\text{K}) = \frac{3(8.314\ \text{J})}{(197\times10^{-3}\ \text{kg})\cdot\text{K}} = \boxed{127\ \text{J/kg}\cdot\text{K}}$$

This agrees with the tabulated value of 129 J/kg · °C within 2%.

P21.51 **(i)** (a) $P_f = \boxed{100\ \text{kPa}}$

(b) $V_f = \dfrac{nRT_f}{P_f}$

$$= \frac{2.00\ \text{mol}(8.314\ \text{J/mol}\cdot\text{K})(400\ \text{K})}{100\times10^{3}\ \text{Pa}} = 0.066\ 5\ \text{m}^3 = \boxed{66.5\ \text{L}}$$

(c) $T_f = \boxed{400\ \text{K}}$

(d) $\Delta E_{\text{int}} = \dfrac{7}{2}nR\Delta T = \dfrac{7}{2}(2.00\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(100\ \text{K})$

$$= \boxed{5.82\ \text{kJ}}$$

(e) $Q = nC_p\Delta T = \dfrac{9}{2}(2.00\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(100\ \text{K})$

$$= \boxed{7.48\ \text{kJ}}$$

(f) $W = -P\Delta V = -nR\Delta T = -(2.00\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(100\ \text{K})$

$$= \boxed{-1.66\ \text{kJ}}$$

(ii) (a) For an isovolumetric process:

$$\frac{P_f}{T_f} = \frac{P_i}{T_i}$$

$$\rightarrow P_f = P_i\frac{T_f}{T_i} = (1.00\times10^{5}\ \text{Pa})\left(\frac{400\ \text{K}}{300\ \text{K}}\right) = \boxed{133\ \text{kPa}}$$

(b) $V_f = V_i = \dfrac{nRT_i}{P_i}$

$$= \frac{(2.00\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{100\times10^{3}\ \text{Pa}} = 0.049\ 9\ \text{m}^3$$

$$= \boxed{49.9\ \text{L}}$$

(c) $T_f = \boxed{400\ \text{K}}$

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(d) $\Delta E_{\text{int}} = \frac{7}{2} nR\Delta T = \boxed{5.82 \text{ kJ}}$ as in (**i**) part (d).

(e) $Q = nC_V \Delta T = \frac{7}{2} nR\Delta T = \frac{7}{2}(2.00 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(100 \text{ K})$

$= \boxed{5.82 \text{ kJ}}$

(f) $W = -\int P dV = \boxed{0}$ since V = constant

(**iii**) (a) $P_f = \boxed{120 \text{ kPa}}$

(b) $V_f = V_i \left(\frac{P_i}{P_f} \right) = (49.9 \text{ L}) \left(\frac{100 \text{ kPa}}{120 \text{ kPa}} \right) = \boxed{41.6 \text{ L}}$

(c) $T_f = T_i = \boxed{300 \text{ K}}$

(d) $\Delta E_{\text{int}} = \frac{7}{2} nR\Delta T = \boxed{0}$ since T = constant

(e) From (f), $Q = \Delta E_{\text{int}} - W = 0 - 909 \text{ J} = \boxed{-909 \text{ J}}$

(f) $W = -\int P dV = -nRT_i \int_{V_i}^{V_f} \frac{dV}{V} = -nRT_i \ln \left(\frac{V_f}{V_i} \right) = -nRT_i \ln \left(\frac{P_i}{P_f} \right)$

$W = -(2.00 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(300 \text{ K}) \ln \left(\frac{100 \text{ kPa}}{120 \text{ kPa}} \right)$

$= \boxed{+909 \text{ J}}$

(**iv**) (a) $P_f = \boxed{120 \text{ kPa}}$

(b) $\gamma = \frac{C_P}{C_V} = \frac{C_V + R}{C_V} = \frac{\frac{7}{2}R + R}{\frac{7}{2}R} = \frac{\frac{9}{2}}{\frac{7}{2}} = \frac{9}{7}$

$P_f V_f^{\gamma} = P_i V_i^{\gamma}$, so

$V_f = V_i \left(\frac{P_i}{P_f} \right)^{1/\gamma} = (49.9 \text{ L}) \left(\frac{100 \text{ kPa}}{120 \text{ kPa}} \right)^{7/9} = \boxed{43.3 \text{ L}}$

(c) $PV = nRT \rightarrow \frac{T_i}{P_i V_i} = \frac{T_f}{P_f V_f}$

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$$\rightarrow T_f = T_i\left(\frac{P_f V_f}{P_i V_i}\right) = (300\text{ K})\left(\frac{120\text{ kPa}}{100\text{ kPa}}\right)\left(\frac{43.3\text{ L}}{49.9\text{ L}}\right) = \boxed{312\text{ K}}$$

(d) $\Delta E_{\text{int}} = \frac{7}{2}nR\Delta T = \frac{7}{2}(2.00\text{ mol})(8.314\text{ J/mol}\cdot\text{K})(12.4\text{ K})$

$= \boxed{+722\text{ J}}$

(e) $Q = \boxed{0}$

(f) $W = -Q + \Delta E_{\text{int}} = 0 + 722\text{ J} = \boxed{+722\text{ J}}$

P21.52 (a) The $\boxed{\text{pressure increases as volume decreases}}$ (and vice versa), so dV/dP is always negative.

In equation form, $\frac{dV}{dP} < 0$ and $-\left(\frac{1}{V}\right)\left(\frac{dV}{dP}\right) > 0$

(b) For an ideal gas, $V = \frac{nRT}{P}$ and $\kappa_1 = -\frac{1}{V}\frac{d}{dP}\left(\frac{nRT}{P}\right)$

For isothermal compression, T is constant and the derivative gives us

$$\kappa_1 = -\frac{nRT}{V}\left(\frac{-1}{P^2}\right) = \frac{1}{P}$$

(c) For an adiabatic compression, $PV^\gamma = C$ (where C is a constant) and we evaluate dV/dP as follows:

$$\kappa_2 = -\left(\frac{1}{V}\right)\frac{d}{dP}\left(\frac{C}{P}\right)^{1/\gamma} = \left(\frac{1}{V\gamma}\right)\frac{C^{1/\gamma}}{\left(P^{1/\gamma+1}\right)} = \frac{V}{V\gamma P} = \frac{1}{\gamma P}$$

(d) $\kappa_1 = \frac{1}{P} = \frac{1}{(2.00\text{ atm})} = \boxed{0.500\text{ atm}^{-1}}$

(e) For a monatomic ideal gas, $\gamma = \frac{C_P}{C_V} = \frac{5}{3}$, so

$$\kappa_2 = \frac{1}{\gamma P} = \frac{1}{\frac{5}{3}(2.00\text{ atm})} = \boxed{0.300\text{ atm}^{-1}}$$

P21.53 The pressure of the gas in the lungs of the diver must be the same as the absolute pressure of the water at this depth of 50.0 meters. This is:

$$P = P_0 + \rho g h$$
$$= 1.00\text{ atm} + \left(1.03\times10^3\text{ kg/m}^3\right)\left(9.80\text{ m/s}^2\right)(50.0\text{ m})$$

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or $$P = 1.00 \text{ atm} + 5.05 \times 10^5 \text{ Pa}\left(\frac{1.00 \text{ atm}}{1.013 \times 10^5 \text{ Pa}}\right) = 5.98 \text{ atm}$$

If the partial pressure due to the oxygen in the gas mixture is to be 1.00 atmosphere (or the fraction $\frac{1}{5.98}$ of the total pressure), oxygen molecules should make up only $\frac{1}{5.98}$ of the total number of molecules. This will be true if 1.00 mole of oxygen is used for every 4.98 mole of helium. The ratio by weight is then

$$\frac{(4.98 \text{ mol He})(4.003 \text{ g/mol He})g}{(1.00 \text{ mol } O_2)(2 \times 15.999 \text{ g/mol } O_2)g} = \boxed{0.623}$$

P21.54 Sulfur dioxide is the gas with the greatest molecular mass of those listed. If the effective spring constants for various chemical bonds are comparable, SO_2 can then be expected to have low frequencies of atomic vibration. Vibration can be excited at lower temperature than for the other gases. Some vibration may be going on at 300 K. With more degrees of freedom for molecular motion, the material has higher specific heat.

P21.55 $$n = \frac{m}{M} = \frac{1.20 \text{ kg}}{0.028\,9 \text{ kg/mol}} = 41.5 \text{ mol}$$

(a) $$V_i = \frac{nRT_i}{P_i} = \frac{(41.5 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(298 \text{ K})}{200 \times 10^3 \text{ Pa}} = \boxed{0.514 \text{ m}^3}$$

(b) $$\frac{P_f}{P_i} = \frac{\sqrt{V_f}}{\sqrt{V_i}} \quad \text{so} \quad V_f = V_i\left(\frac{P_f}{P_i}\right)^2 = (0.514 \text{ m}^3)\left(\frac{400}{200}\right)^2 = \boxed{2.06 \text{ m}^3}$$

(c) $$T_f = \frac{P_f V_f}{nR} = \frac{(400 \times 10^3 \text{ Pa})(2.06 \text{ m}^3)}{(41.5 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})} = \boxed{2.38 \times 10^3 \text{ K}}$$

(d) $$W = -\int_{V_i}^{V_f} P\,dV = -C\int_{V_i}^{V_f} V^{1/2}\,dV = -\left(\frac{P_i}{V_i^{1/2}}\right)\frac{2V^{3/2}}{3}\Bigg|_{V_i}^{V_f}$$

$$= -\frac{2}{3}\left(\frac{P_i}{V_i^{1/2}}\right)\left(V_f^{3/2} - V_i^{3/2}\right)$$

$$W = -\frac{2}{3}\left(\frac{200 \times 10^3 \text{ Pa}}{\sqrt{0.514 \text{ m}^3}}\right)\left[(2.06 \text{ m}^3)^{3/2} - (0.514 \text{ m}^3)^{3/2}\right] = \boxed{-480 \text{ kJ}}$$

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(e) $\Delta E_{\text{int}} = nC_V\Delta T = (41.5\text{ mol})\left[\frac{5}{2}(8.314\text{ J/mol}\cdot\text{K})\right](2.38\times10^3 - 298)\text{ K}$

$= 1.80\times10^6\text{ J}$

$Q = \Delta E_{\text{int}} - W = 1.80\times10^6\text{ J} + 4.80\times10^5\text{ J} = 2.28\times10^6\text{ J} = \boxed{2.28\text{ MJ}}$

P21.56 (a) Begin with Equation 17.8 and substitute the definition of bulk modulus from Equation 12.8:

$$v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{1}{\rho}\left(-V\frac{dP}{dV}\right)}$$

Now substitute using Equation 21.37 with the constant on the right hand side represented by K:

$$v = \sqrt{\frac{-V}{\rho}\frac{d}{dV}\left(KV^{-\gamma}\right)} = \sqrt{\frac{-KV}{\rho}\left(-\gamma V^{-\gamma-1}\right)} = \sqrt{\frac{\gamma}{\rho}\left(KV^{-\gamma}\right)}$$

$$= \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma}{\rho}\frac{nRT}{V}} = \sqrt{\frac{\gamma(m/M)RT}{m}} = \sqrt{\frac{\gamma RT}{M}}$$

(b) $v = \sqrt{\dfrac{1.40(8.314\text{ J/mol}\cdot\text{K})(293\text{ K})}{0.028\,9\text{ kg/mol}}} = \boxed{344\text{ m/s}}$

This agrees within 0.2% with the 343 m/s listed in Table 17.1.

(c) We use $k_B = \dfrac{R}{N_A}$ and $M = m_0N_A$: $v = \sqrt{\dfrac{\gamma RT}{M}} = \sqrt{\dfrac{\gamma k_BN_AT}{m_0N_A}} = \sqrt{\dfrac{\gamma k_BT}{m_0}}$

(d) The most probable molecular speed is $\sqrt{\dfrac{2k_BT}{m_0}}$, the average speed is $\sqrt{\dfrac{8k_BT}{\pi m_0}}$, and the rms speed is $\sqrt{\dfrac{3k_BT}{m_0}}$.

The speed of sound is somewhat less than each measure of molecular speed. Sound propagation is orderly motion overlaid on the disorder of molecular motion.

P21.57 (a) The average speed v_{avg} is just the weighted average of all the speeds.

$$v_{\text{avg}} = \frac{2(v)+3(2v)+5(3v)+4(4v)+3(5v)+2(6v)+1(7v)}{2+3+5+4+3+2+1}$$

$$= \boxed{3.65v}$$

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(b) First find the average of the square of the speeds,

$$(v^2)_{\text{avg}} = \frac{2(v)^2 + 3(2v)^2 + 5(3v)^2 + 4(4v)^2 + 3(5v)^2 + 2(6v)^2 + 1(7v)^2}{2+3+5+4+3+2+1}$$

$$= 15.95v^2$$

The root-mean square speed is then $v_{\text{rms}} = \sqrt{(v_{avg})^2} = \boxed{3.99v}$

(c) The most probable speed is the one that most of the particles have; i.e., five particles have speed $\boxed{3.00v}$.

(d) $PV = \frac{1}{3} N m_0 v_{\text{av}}^2$

Therefore, $P = \frac{20}{3}\frac{\left[m_0(15.95)v^2\right]}{V} = \boxed{106\left(\frac{m_0 v^2}{V}\right)}$

(e) The average kinetic energy for each particle is

$$\bar{K} = \frac{1}{2} m_0 v_{\text{av}}^2 = \frac{1}{2} m_0 \left(15.95v^2\right) = \boxed{7.98 m_0 v^2}$$

P21.58 (a) For the adiabatic process $PV^\gamma = k$, a constant. The work is

$$W = -\int_i^f P\,dV = -k\int_{V_i}^{V_f} \frac{dV}{V^\gamma} = \left.\frac{-kV^{1-\gamma}}{1-\gamma}\right|_{V_i}^{V_f}$$

For k we can substitute $P_i V_i^\gamma$ and also $P_f V_f^\gamma$ to have

$$W = -\frac{P_f V_f^\gamma V_f^{1-\gamma} - P_i V_i^\gamma V_i^{1-\gamma}}{1-\gamma} = \frac{P_f V_f - P_i V_i}{\gamma - 1}$$

(b) For an adiabatic process $\Delta E_{\text{int}} = Q + W$ and $Q = 0$. Therefore,

$$W = \Delta E_{\text{int}} = nC_V \Delta T = nC_V\left(T_f - T_i\right)$$

. (c) The expressions are equal because $PV = nRT$ and $\gamma = (C_V + R)/C_V = 1 + R/C_V$ give $R = (\gamma - 1)C_V$, so $PV = n(\gamma - 1)C_V T$ and $PV/(\gamma - 1) = nC_V T$.

P21.59 (a) $\Delta E_{\text{int}} = Q + W = 0 + W \quad \rightarrow \quad W = nC_V\left(T_f - T_i\right)$

$$-2\ 500\ \text{J} = (1\ \text{mol})\left(\frac{3}{2}\right)\left(8.314\ \text{J/mol}\cdot\text{K}\right)\left(T_f - 500\ \text{K}\right)$$

$$T_f = \boxed{300\ \text{K}}$$

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(b) $P_i V_i^{\gamma} = P_f V_f^{\gamma}$

$$P_i\left(\frac{nRT_i}{P_i}\right)^{\gamma} = P_f\left(\frac{nRT_f}{P_f}\right)^{\gamma}$$

$$T_i^{\gamma} P_i^{1-\gamma} = T_f^{\gamma} P_f^{1-\gamma}$$

$$\frac{T_i^{\gamma/(\gamma-1)}}{P_i} = \frac{T_f^{\gamma/(\gamma-1)}}{P_f}$$

$$P_f = P_i\left(\frac{T_f}{T_i}\right)^{\gamma/(\gamma-1)}$$

$$P_f = P_i\left(\frac{T_f}{T_i}\right)^{(5/3)(3/2)} = (3.60\text{ atm})\left(\frac{300}{500}\right)^{5/2} = \boxed{1.00\text{ atm}}$$

P21.60 (a) The process is adiabatic:

$$\Delta E_{\text{int}} = Q + W = 0 + W \quad \rightarrow \quad W = nC_V\left(T_f - T_i\right)$$

For an ideal gas,

$$W = nC_V\left(T_f - T_i\right) = n\left(\frac{3}{2}\right)R\left(T_f - T_i\right)$$

Solving for final temperature, we get

$$T_f = \boxed{T_i + \frac{2}{3}\frac{W}{nR}}$$

(b) $P_i V_i^{\gamma} = P_f V_f^{\gamma}$

$$P_i\left(\frac{nRT_i}{P_i}\right)^{\gamma} = P_f\left(\frac{nRT_f}{P_f}\right)^{\gamma} \quad \rightarrow \quad T_i^{\gamma} P_i^{1-\gamma} = T_f^{\gamma} P_f^{1-\gamma}$$

$$\rightarrow \quad \frac{T_i^{\gamma/(\gamma-1)}}{P_i} = \frac{T_f^{\gamma/(\gamma-1)}}{P_f} \quad \rightarrow \quad P_f = P_i\left(\frac{T_f}{T_i}\right)^{\gamma/(\gamma-1)}$$

where $\gamma = \dfrac{C_p}{C_V} = \dfrac{5}{3}$ for an ideal gas, and $\dfrac{\gamma}{\gamma-1} = \dfrac{\frac{5}{3}}{\frac{5}{3}-1} = \dfrac{\frac{5}{3}}{\frac{2}{3}} = \dfrac{5}{2}$.

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Substituting this and the result from part (a) gives

$$P_f = P_i\left(\frac{T_i + \frac{2}{3}\frac{W}{nR}}{T_i}\right)^{5/2} = \boxed{P_i\left(1 + \frac{2}{3}\frac{W}{nRT_i}\right)^{5/2}}$$

***P21.61** (a) Let $d = 2r$ represent the diameter of the particle. Its mass is $m = \rho V = \rho\frac{4}{3}\pi r^3 = \rho\frac{4}{3}\pi\left(\frac{d}{2}\right)^3 = \frac{\rho\pi d^3}{6}$. Then $\frac{1}{2}mv_{\text{rms}}^2 = \frac{3}{2}kT$ gives

$$\frac{\rho\pi d^3}{6}v_{\text{rms}}^2 = 3kT$$

so

$$v_{\text{rms}} = \left(\frac{18kT}{\rho\pi d^3}\right)^{1/2} = \left(\frac{18(1.38\times10^{-23}\text{ J/K})(293\text{ K})}{\pi(1\,000\text{ kg/m}^3)}\right)^{1/2} d^{-3/2}$$

$$= \boxed{(4.81\times10^{-12})d^{-3/2}}$$

where v_{rms} is in meters per second and d is in meters.

(b) $v = d/\Delta t \rightarrow (4.81\times10^{-12}\text{ m}^{5/2}/\text{s})d^{-3/2} = d/\Delta t$

$$\Delta t = \frac{d}{(4.81\times10^{-12}\text{ m}^{5/2}/\text{s})d^{-3/2}}$$

$$= \boxed{(2.08\times10^{11})d^{5/2}}$$

where Δt is in seconds and d is in meters.

(c) $$v_{\text{rms}} = \left(\frac{18kT}{\rho\pi d^3}\right)^{1/2} = \left(\frac{18(1.38\times10^{-23}\text{ J/K})(293\text{ K})}{(1\,000\text{ kg/m}^3)\pi(3\times10^{-6}\text{ m})^3}\right)^{1/2}$$

$$= \boxed{0.926\text{ mm/s}}$$

$$v = \frac{x}{\Delta t} \rightarrow \Delta t = \frac{x}{v} = \frac{3\times10^{-6}\text{ m}}{9.26\times10^{-4}\text{ m/s}} = \boxed{3.24\text{ ms}}$$

(d) $70\text{ kg} = (1\,000\text{ kg/m}^3)\frac{\pi d^3}{6} \rightarrow d = 0.511\text{ m}$

$$v_{\text{rms}} = \left(\frac{18kT}{\rho\pi d^3}\right)^{1/2} = \left(\frac{18(1.38\times10^{-23}\text{ J/K})(293\text{ K})}{(1\,000\text{ kg/m}^3)\pi(0.511\text{ m})^3}\right)^{1/2}$$

$$= \boxed{1.32\times10^{-11}\text{ m/s}}$$

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$$\Delta t = \frac{0.511 \text{ m}}{1.32 \times 10^{-11} \text{ m/s}} = \boxed{3.88 \times 10^{10} \text{ s}} = 1\,230 \text{ yr}$$

This motion is too slow to observe.

P21.62 (a) Maxwell's speed distribution function is

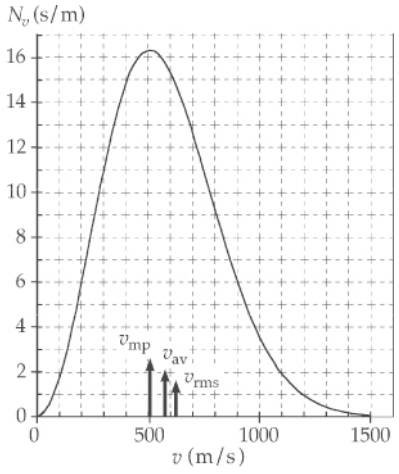
$$N_v = 4\pi N \left(\frac{m_0}{2\pi k_B T} \right)^{3/2} v^2 e^{-m_0 v^2 / 2k_B T}$$

With $N = 1.00 \times 10^4$, $m_0 = \frac{M}{N_A} = \frac{0.032 \text{ kg}}{6.02 \times 10^{23}} = 5.32 \times 10^{-26}$ kg,

$T = 500$ K, and $k_B = 1.38 \times 10^{-23}$ J/molecule · K, this becomes

$$N_v = \left(1.71 \times 10^{-4}\right) v^2 e^{-\left(3.85 \times 10^{-6}\right) v^2}$$

ANS. FIG. P21.62(a) below is a plot of this function for the range $0 \le v \le 1\,500$ m/s.



ANS. FIG. P21.62(a)

(b) The most probable speed occurs where N_v is a maximum.

From the graph, $\boxed{v_{mp} \approx 510 \text{ m/s}}$.

(c) $$v_{avg} = \sqrt{\frac{8k_B T}{\pi m_0}}$$

$$= \sqrt{\frac{8\left(1.38 \times 10^{-23} \text{ J/molecule} \cdot \text{K}\right)(500 \text{ K})}{\pi\left(5.32 \times 10^{-26} \text{ kg}\right)}} = \boxed{575 \text{ m/s}}$$

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Also,

$$v_{\text{rms}} = \sqrt{\frac{3k_BT}{m_0}} = \sqrt{\frac{3\left(1.38\times10^{-23}\ \text{J/molecule}\cdot\text{K}\right)(500\ \text{K})}{5.32\times10^{-26}\ \text{kg}}}$$

$$= \boxed{624\ \text{m/s}}$$

(d) The fraction of particles in the range $300\ \text{m/s} \le v \le 600\ \text{m/s}$ is

$$\frac{\int_{300}^{600} N_v\, dv}{N}$$

where $N = 10^4$ and the integral of N_v is read from the graph as the area under the curve. This is approximately the area of a large rectangle 11 s/m high and 300 m/s wide [corners at (300, 0), (300, 11), (600, 11), and (600, 0)], plus a smaller rectangle 5.5 s/m high and 100 m/s wide [corners at (500, 11), (500, 16.6), (600, 16.5), and (600, 11)], plus a triangle 5.5 s/m high with a 200 m/s base [vertices at (300, 11), (500, 16.5), and (500, 11)]:

$$(11)(300) + (5.5)(100) + (1/2)(5.5)(200) = 4\ 400$$

and the fraction is 0.44 or $\boxed{44\%}$.

P21.63 For the system of the ball and the air, Equation 8.2 gives us,

$$\Delta K + \Delta E_{\text{int}} = 0$$

Substitute for the internal energy and solve for the temperature increase of the air:

$$\Delta K + nC_V\Delta T = 0 \ \rightarrow\ \Delta T = \frac{-\Delta K}{nC_V} = \frac{-\Delta K}{(m/M)\left(\frac{5}{2}R\right)} = -\frac{2M\Delta K}{5mR}$$

Express the mass m of the air in terms of the density and volume of the cylinder through which the ball passes, and evaluate the change in kinetic energy of the ball:

$$\Delta T = -\frac{2M\Delta K}{5\rho VR} = -\frac{2M\left(\frac{1}{2}m_{\text{ball}}v_f^2 - \frac{1}{2}m_{\text{ball}}v_i^2\right)}{5\rho\left(\pi r^2\ell\right)R}$$

$$= \frac{Mm_{\text{ball}}\left(v_i^2 - v_f^2\right)}{5\pi\rho r^2\ell R}$$

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Substitute numerical values:

$$\Delta T = \frac{(28.9 \times 10^{-3}\ \text{kg/mol})(0.142\ \text{kg})\left[(47.2\ \text{m/s})^2 - (42.5\ \text{m/s})^2\right]}{5\pi(1.20\ \text{kg/m}^3)(0.037\ 0\ \text{m})^2(16.8\ \text{m})(8.314\ \text{J/mol}\cdot°\text{C})}$$

$$= \boxed{0.480°\text{C}}$$

P21.64 (a) The latent heat of evaporation per molecule is

$$2\ 430\ \text{J/g} = (2430\ \text{J/g})\left(\frac{18.0\ \text{g}}{1\ \text{mol}}\right)\left(\frac{1\ \text{mol}}{6.02\times10^{23}\ \text{molecule}}\right)$$

$$= \boxed{7.27\times10^{-20}\ \text{J/molecule}}$$

If the molecule is about to break free, we assume that it possesses the energy as translational kinetic energy.

(b) Consider one gram of these molecules. From $K = \frac{1}{2}mv^2$ we obtain

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(2\ 430\ \text{J})}{10^{-3}\ \text{kg}}} = 2.20\times10^3\ \text{m/s} = \boxed{2.20\ \text{km/s}}$$

(c) The total translational kinetic energy of an ideal gas is $\frac{3}{2}nrT$, so we have

$$(2\ 430\ \text{J/g})\left(\frac{18.0\ \text{g}}{1\ \text{mol}}\right) = \frac{3}{2}(1\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})T$$

which gives

$$T = \boxed{3.51\times10^3\ \text{K}}$$

(d) The evaporating particles emerge with much less kinetic energy, as negative work is performed on them by restraining forces as they leave the liquid. Much of the initial kinetic energy is used up in overcoming the latent heat of vaporization. There are also very few of these escaping at any moment in time.

P21.65 (a) $$n = \frac{PV}{RT} = \frac{(1.013\times10^5\ \text{Pa})(5.00\times10^{-3}\ \text{m}^3)}{(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K})}$$

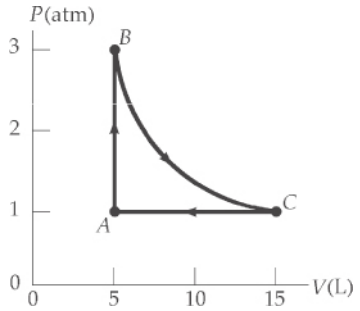
$$= \boxed{0.203\ \text{mol}}$$

(b) $$T_B = T_A\left(\frac{P_B}{P_A}\right) = (300\ \text{K})\left(\frac{3.00}{1.00}\right) = \boxed{900\ \text{K}}$$

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(c) $T_C = T_B = \boxed{900\ \text{K}}$

(d) $V_C = V_A\left(\frac{T_C}{T_A}\right) = (5.00\ \text{L})\left(\frac{900}{300}\right) = \boxed{15.0\ \text{L}}$



ANS. FIG. P21.65

(e) $A \to B$: lock the piston in place and put the cylinder into an oven at 900 K, gradually heating the gas. $B \to C$: keep the sample in the oven while gradually letting the gas expand to lift a load on the piston as far as it can. $C \to A$: carry the cylinder back into the room at 300 K and let the gas gradually cool and contract without touching the piston.

(f) For $A \to B$: $W = \boxed{0}$

$$\Delta E_{\text{int}} = nC_V\Delta T = n\left(\frac{3}{2}\right)R\Delta T = Q + W$$

$$\Delta E_{\text{int}} = (0.203\ \text{mol})\left(\frac{3}{2}\right)(8.314\ \text{J/mol}\cdot\text{K})(600\ \text{K}) = \boxed{1.52\ \text{kJ}}$$

$$Q = \Delta E_{\text{int}} - W = \boxed{1.52\ \text{kJ}}$$

For $B \to C$: $\Delta E_{\text{int}} = \boxed{0}$, because $\Delta T = 0$;

$$W = -nRT_B \ln\left(\frac{V_C}{V_B}\right)$$

$$W = -(0.203\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(900\ \text{K})\ln(3.00)$$

$$= \boxed{-1.67\ \text{kJ}}$$

$$Q = \Delta E_{\text{int}} - W = \boxed{1.67\ \text{kJ}}$$

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For $C \to A$: $\Delta E_{\text{int}} = nC_V \Delta T$

$$\Delta E_{\text{int}} = (0.203 \text{ mol})\left(\frac{3}{2}\right)(8.314 \text{ J/mol} \cdot \text{K})(-600 \text{ K})\text{J}$$

$$= \boxed{-1.52 \text{ kJ}}$$

$$W = -P\Delta V = -nR\Delta T$$

$$= -(0.203 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(-600 \text{ K})$$

$$= \boxed{1.01 \text{ kJ}}$$

$$Q = \Delta E_{\text{int}} - W = -1.52 \text{ kJ} - 1.01 \text{ kJ} = \boxed{-2.53 \text{ kJ}}$$

(g) We add the amounts of energy for each process to find them for the whole cycle.

$$Q_{ABCA} = +1.52 \text{ kJ} + 1.67 \text{ kJ} - 2.53 \text{ kJ} = \boxed{0.656 \text{ kJ}}$$

$$W_{ABCA} = 0 - 1.67 \text{ kJ} + 1.01 \text{ kJ} = \boxed{-0.656 \text{ kJ}}$$

$$(\Delta E_{\text{int}})_{ABCA} = Q_{ABCA} + W_{ABCA} = +1.52 \text{ kJ} + 0 - 1.52 \text{ kJ} = \boxed{0}$$

For any cyclic process, $\Delta E_{\text{int}} = 0$.

P21.66 (a) The effect of large centripetal acceleration is like the effect of a very high gravitational field on an atmosphere. The result is:

The larger-mass molecules settle to the outside while the region at smaller r has a higher concentration of low-mass molecules.

(b) Consider a single kind of molecule, all of mass m_0. To cause the centripetal acceleration of the molecules between r and $r + dr$, the inward force must increase with increasing distance from the center according to $\sum F_r = m_0 a_r$. Taking the positive direction toward the center of the centrifuge, we have

$$(P + dP)A - PA = n_V (m_0 A\, dr)(r\omega^2)$$

where $n_V = n_V(r) = N/V$, the number of molecules per unit volume, is an implicit function of r, and A is the area of any cylindrical shell of thickness dr and radius r. The equation reduces to

$$dP = n_V m_0 \omega^2 r dr \qquad \textbf{[1]}$$

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But also within any small cylindrical shell,

$$PV = Nk_BT \rightarrow P = \left(\frac{N}{V}\right)k_BT$$

$$\rightarrow dP = d\left(\frac{N}{V}\right)k_BT = d\left(n_V\right)k_BT = dn_Vk_BT$$

Therefore, equation [1] becomes

$$dn_Vk_BT = n_Vm_0\omega^2 rdr \quad \rightarrow \quad \frac{dn_V}{n_V} = \frac{m_0\omega^2}{k_BT}rdr$$

giving $\int_{n_0}^{n}\frac{dn_V}{n_V} = \frac{m_0\omega^2}{k_BT}\int_0^r rdr$, where $n_V\left(r=0\right) = n_0$.

Integrating, we find

$$\ln\left(n_V\right)\Big|_{n_0}^{n_V} = \frac{m_0\omega^2}{k_BT}\left(\frac{r^2}{2}\right)\Big|_0^r \quad \rightarrow \quad \ln\left(\frac{n_V}{n_0}\right) = \frac{m_0\omega^2}{2k_BT}r^2$$

and solving for $n \equiv n_V$, we have $\boxed{n = n_0e^{m_0r^2\omega^2/2k_BT}}$.

P21.67 $N_v\left(v\right) = 4\pi N\left(\frac{m_0}{2\pi k_BT}\right)^{3/2}v^2\exp\left(\frac{-m_0v^2}{2k_BT}\right)$, where $\exp(x)$ represents e^x

Note that $v_{\text{mp}} = \left(\frac{2k_BT}{m_0}\right)^{1/2}$.

Thus, $N_v\left(v\right) = 4\pi N\left(\frac{m_0}{2\pi k_BT}\right)^{3/2}v^2e^{\left(-v^2/v_{\text{mp}}^2\right)}$

and $\frac{N_v\left(v\right)}{N_v\left(v_{\text{mp}}\right)} = \left(\frac{v}{v_{\text{mp}}}\right)^2 e^{\left(1-v^2/v_{\text{mp}}^2\right)}$.

For $v = \frac{v_{\text{mp}}}{50}$,

$$\frac{N_v\left(v\right)}{N_v\left(v_{\text{mp}}\right)} = \left(\frac{1}{50}\right)^2 e^{\left[1-(1/50)^2\right]} = 1.09\times10^{-3}$$

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The other values are computed similarly, with the following results:

	$\frac{v}{v_{\text{mp}}}$	$\frac{N_v(v)}{N_v\left(v_{\text{mp}}\right)}$
(a)	$\frac{1}{50}$	1.09×10^{-3}
(b)	$\frac{1}{10}$	2.69×10^{-2}
(c)	$\frac{1}{2}$	0.529
(d)	1	1.00
(e)	2	0.199
(f)	10	1.01×10^{-41}
(g)	50	$1.25 \times 10^{-1\,082}$

To find the last value, we note:

$$(50)^2 e^{1-2\,500} = 2\,500e^{-2\,499}$$

$$10^{\log 2\,500} e^{(\ln 10)(-2\,499/\ln 10)} = 10^{\log 2\,500} 10^{-2\,499/\ln 10} = 10^{\log 2\,500 - 2\,499/\ln 10}$$

$$= 10^{-1\,081.904} = 10^{0.096} \times 10^{-1\,082}$$

P21.68 (a) The energy of one molecule can be represented as

$$\frac{1}{2}m_0 v_x^2 + \frac{1}{2}m_0 v_y^2 + \frac{1}{2}m_0 v_z^2 + \frac{1}{2}I\omega_x^2 + \frac{1}{2}I\omega_z^2$$

Its average value is

$$\frac{1}{2}k_B T + \frac{1}{2}k_B T + \frac{1}{2}k_B T + \frac{1}{2}k_B T + \frac{1}{2}k_B T = \frac{5}{2}k_B T$$

The energy of one mole is obtained by multiplying by Avogadro's number, $E_{\text{int}} / n = \frac{5}{2}RT$.

And the molar heat capacity at constant volume is

$$E_{\text{int}} / nT = \boxed{\frac{5}{2}R}.$$

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(b) The energy of one molecule can be represented as

$$\frac{1}{2}m_0v_x^2+\frac{1}{2}m_0v_y^2+\frac{1}{2}m_0v_z^2+\frac{1}{2}I\omega_x^2+\frac{1}{2}I\omega_z^2+\frac{1}{2}I\omega_y^2$$

Its average value is

$$\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT=3k_BT$$

The energy of one mole is obtained by multiplying by Avogadro's number, $E_{\text{int}}/n = 3RT$.

And the molar heat capacity at constant volume is $E_{\text{int}}/nT = \boxed{3R}$.

(c) Let the modes of vibration be denoted by 1 and 2. The energy of one molecule can be represented as

$$\frac{1}{2}m_0\left(v_x^2+{}_0\,v_y^2+v_z^2\right)+\frac{1}{2}I\omega_x^2+\frac{1}{2}I\omega_z^2 +\left(\frac{1}{2}\mu v_{\text{rel}}^2+\frac{1}{2}kx^2\right)_1+\left(\frac{1}{2}\mu v_{\text{rel}}^2+\frac{1}{2}kx^2\right)_2$$

Its average value is

$$\frac{3}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT=\frac{9}{2}k_BT$$

The energy of one mole is obtained by multiplying by Avogadro's number, $E_{\text{int}}/n=\frac{9}{2}RT$.

And the molar heat capacity at constant volume is

$E_{\text{int}}/nT=\boxed{\frac{9}{2}R}$.

(d) The energy of one molecule can be represented as

$$\frac{1}{2}m_0\left(v_x^2+{}_0\,v_y^2+v_z^2\right)+\frac{1}{2}I\omega_x^2+\frac{1}{2}I\omega_z^2+\frac{1}{2}I\omega_y^2 +\left(\frac{1}{2}\mu v_{\text{rel}}^2+\frac{1}{2}kx^2\right)_1+\left(\frac{1}{2}\mu v_{\text{rel}}^2+\frac{1}{2}kx^2\right)_2$$

Its average value is

$$\frac{3}{2}k_BT+\frac{3}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT+\frac{1}{2}k_BT=5k_BT$$

The energy of one mole is obtained by multiplying by Avogadro's number, $E_{\text{int}}/\text{n} = 5RT$.

And the molar heat capacity at constant volume is $E_{\text{int}}/\text{n}T = \boxed{5R}$.

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(e) Measure the constant-volume specific heat of the gas as a function of temperature and look for plateaus on the graph. If the first jump goes from $\frac{3}{2}R$ to $\frac{5}{2}R$, the molecules can be diagnosed as linear. If the first jump goes from $\frac{3}{2}R$ to $3R$, the molecules must be nonlinear. The tabulated data at one temperature are insufficient for the determination. At room temperature some of the heavier molecules appear to be vibrating.

P21.69 (a) First find $\overline{v^2}$ as $\overline{v^2} = \frac{1}{N}\int_0^\infty v^2 N_v\, dv$. Let $a = \frac{m_0}{2k_B T}$.

Then, $\overline{v^2} = \frac{\left[4N\pi^{-1/2}a^{3/2}\right]}{N}\int_0^\infty v^4 e^{-av^2 dv} = \left[4a^{3/2}\pi^{-1/2}\right]\frac{3}{8a^2}\sqrt{\frac{\pi}{a}} = \frac{3k_B T}{m}$

The root-mean square speed is then $v_{\text{rms}} = \sqrt{\overline{v^2}} = \boxed{\sqrt{\frac{3k_B T}{m_0}}}$

(b) To find the average speed, we have

$$v_{\text{avg}} = \frac{1}{N}\int_0^\infty v N_v\, dv = \frac{\left(4Na^{3/2}\pi^{-1/2}\right)}{N}\int_0^\infty v^3 e^{-av^2}\, dv = \frac{4a^{3/2}\pi^{-1/2}}{2a^2}$$

$$= \boxed{\sqrt{\frac{8k_B T}{\pi m_0}}}$$

P21.70 We want to evaluate $\frac{dP}{dV}$ for the function implied by $PV = nRT$ = constant, and also for the different function implied by PV^γ = constant. We can use implicit differentiation:

From PV = constant $\quad P\frac{dV}{dV} + V\frac{dP}{dV} = 0 \quad \rightarrow \quad \left(\frac{dP}{dV}\right)_{\text{isotherm}} = -\frac{P}{V}$

From PV^γ = constant $\quad P\gamma V^{\gamma-1} + V^\gamma \frac{dP}{dV} = 0 \quad \rightarrow \quad \left(\frac{dP}{dV}\right)_{\text{adiabat}} = -\frac{\gamma P}{V}$

Therefore, $\quad \left(\frac{dP}{dV}\right)_{\text{adiabat}} = \gamma\left(\frac{dP}{dV}\right)_{\text{isotherm}}$

The theorem is proved.

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P21.71 (a) The number of molecules in the pot is given by

$$(10\,000\text{ g})\left(\frac{1.00\text{ mol}}{18.0\text{ g}}\right)\left(\frac{6.02\times10^{23}\text{ molecules}}{1.00\text{ mol}}\right)$$

$$= \boxed{3.34\times10^{26}\text{ molecules}}$$

(b) Each day, $\frac{1}{10}$ of the original molecules are left in the pot. Let us find out how many days are required for there to be one molecule left. We solve the following equation for n_d, the number of days:

$$\text{Number left} = 1 = \left(\tfrac{1}{10}\right)^{n_d} N \quad \rightarrow \quad \frac{1}{N} = \left(\tfrac{1}{10}\right)^{n_d}$$

where N is the original number of molecules. Take the logarithm of both sides:

$$\log\frac{1}{N} = -\log N = \log\left(\tfrac{1}{10}\right)^{n_d} = n_d\log\left(\tfrac{1}{10}\right) = -n_d\log 10 = -n_d$$

$$\rightarrow \quad n_d = \log N$$

Substitute the numerical value for N:

$$n_d = \log\left(3.34\times10^{26}\right) = 26.5$$

Therefore, the last molecule is ladled out after the 26th day and so during the 27th day.

(c) The soup is this fraction of the hydrosphere: $\left(\frac{10.0\text{ kg}}{1.32\times10^{21}\text{ kg}}\right)$

Therefore, today's soup likely contains this fraction of the original molecules. The number of original molecules likely in the pot again today is

$$\left(\frac{10.0\text{ kg}}{1.32\times10^{21}\text{ kg}}\right)\left(3.34\times10^{26}\text{ molecules}\right)$$

$$= \boxed{2.53\times10^{6}\text{ molecules}}$$

P21.72 (a) Consider the molecule-Earth system to be isolated. Treat an escaping molecule as going from $r = R_E$, $v = v_0$, to $r = \infty$, $v = 0$:

$$\Delta K + \Delta U = 0$$

$$\left(0 - \frac{1}{2}m_0v^2\right) + \left[0 - \left(-\frac{Gm_0M}{R_E}\right)\right] = 0 \quad \rightarrow \frac{1}{2}m_0v^2 = \frac{Gm_0M}{R_E}$$

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Since the free-fall acceleration at the surface is, $g = \frac{GM}{R_E^2}$, this can also be written as: $\frac{1}{2}m_0v^2 = \frac{Gm_0M}{R_E} = \boxed{m_0gR_E}$

(b) For O_2, the mass of one molecule is

$$m_0 = \frac{0.0320 \text{ kg/mol}}{6.02\times10^{23} \text{ molecules/mol}} = 5.32\times10^{-26} \text{ kg/molecule}$$

Then, if $m_0gR_E = 10\left(\frac{3k_BT}{2}\right)$, the temperature is

$$T = \frac{m_0gR_E}{15k_B} = \frac{\left(5.32\times10^{-26} \text{ kg}\right)\left(9.80 \text{ m/s}^2\right)\left(6.37\times10^{6} \text{ m}\right)}{15\left(1.38\times10^{-23} \text{ J/mol}\cdot\text{K}\right)}$$

$$= \boxed{1.60\times10^{4} \text{ K}}$$

P21.73 (a) For sodium atoms (with a molar mass M = 23.0 g/mol):

$$\frac{1}{2}m_0v^2 = \frac{3}{2}k_BT$$

$$\frac{1}{2}\left(\frac{M}{N_A}\right)v^2 = \frac{3}{2}k_BT$$

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3\left(8.314 \text{ J/mol}\cdot\text{K}\right)\left(2.40\times10^{-4} \text{ K}\right)}{23.0\times10^{-3} \text{ kg}}}$$

$$= \boxed{0.510 \text{ m/s}}$$

(b) $\Delta t = \frac{d}{v_{rms}} = \frac{0.010\ 0 \text{ m}}{0.510 \text{ m/s}} = 19.6 \text{ ms} \approx \boxed{20 \text{ ms}}$

Challenge Problems

P21.74 (a) The average value of a collection of particle speeds is

$$v_{avg} = \frac{\sum_i^N v_i}{N}$$

Use this equation to find the average for the two speeds given in the problem:

$$v_{avg} = \frac{v_1 + v_2}{2} = \frac{av_{avg} + \left(2-a\right)v_{avg}}{2} = v_{avg}$$

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(b) The rms average value of a collection of particle speeds is

$$v_{\text{rms}} = \frac{\sum_i^N v_i^2}{N}$$

Use this equation to find the square of the rms average for the two speeds given in the problem:

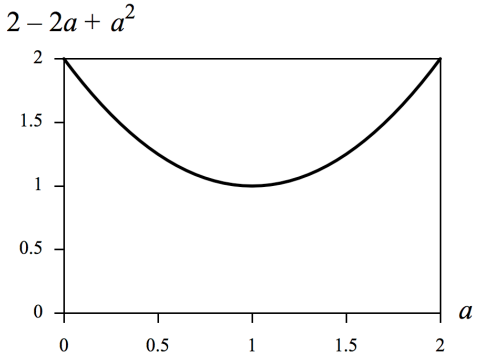
$$v_{\text{rms}}^2 = \frac{v_1^2 + v_2^2}{2} = \frac{\left(av_{\text{avg}}\right)^2 + \left[\left(2-a\right)v_{\text{avg}}\right]^2}{2}$$

$$= \frac{v_{\text{avg}}^2}{2}\left[a^2 + \left(4 - 4a + a^2\right)\right]$$

$$= v_{\text{avg}}^2\left(2 - 2a + a^2\right)$$

$$v_{\text{rms}}^2 = v_{\text{avg}}^2\left(2 - 2a + a^2\right) \qquad \textbf{[1]}$$

(c) The graph of $(2 - 2a + a^2)$ versus a appears below, over the range of possible values $0 \le a \le 2$.



ANS. FIG. P21.74(c)

Because the factor $(2 - 2a + a^2)$ is generally larger than 1, equation [1] tells us that $v_{\text{rms}} > v_{\text{avg}}$ except at one point in the graph.

(d) From the graph, we see that that $v_{\text{rms}} = v_{\text{avg}}$ when the factor $(2 - 2a + a^2) = 1$, which occurs at $\boxed{a = 1}$.

P21.75 Let the subscripts "1" and "2" refer to the hot and cold compartments, respectively. The pressure is higher in the hot compartment, therefore the hot compartment expands and the cold compartment contracts. Because the walls of the cylinder are insulating, the total internal energy of the system must remain constant:

$$\Delta E_{\text{int}} = 0:$$

$$\Delta E_{\text{int},1} + \Delta E_{\text{int},2} = 0$$

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$$nC_V\Delta T_1 + nC_V\Delta T_2 = 0$$
$$\left(T_{1f} - T_{1i}\right) + \left(T_{2f} - T_{2i}\right) = 0$$
$$T_{1f} + T_{2f} = T_{1i} + T_{2i} = 550\text{ K} + 250\text{ K} = 800\text{ K} \quad \textbf{[1]}$$

Consider the adiabatic changes of the gases.

$$P_{1i}V_{1i}^{\gamma} = P_{1f}V_{1f}^{\gamma} \qquad \text{and} \qquad P_{2i}V_{2i}^{\gamma} = P_{2f}V_{2f}^{\gamma}$$

or
$$\frac{P_{1i}V_{1i}^{\gamma}}{P_{2i}V_{2i}^{\gamma}} = \frac{P_{1f}V_{1f}^{\gamma}}{P_{2f}V_{2f}^{\gamma}}$$

The initial volumes are equal: $V_{1i} = V_{2i}$. Applying the particle in equilibrium model to the piston, the final force on each side of the piston must be the same, and the areas on each side are equal, therefore $P_{1f} = P_{2f}$. The equation simplifies to

$$\frac{P_{1i}}{P_{2i}} = \left(\frac{V_{1f}}{V_{2f}}\right)^{\gamma}$$

Using the ideal gas law:

$$\frac{nRT_{1i}/V_{1i}}{nRT_{2i}/V_{2i}} = \left(\frac{nRT_{1f}/P_{1f}}{nRT_{2f}/P_{2f}}\right)^{\gamma}$$

Simplifying, this gives

$$\frac{T_{1i}}{T_{2i}} = \left(\frac{T_{1f}}{T_{2f}}\right)^{\gamma}$$

since $V_{1i} = V_{2i}$ and $P_{1f} = P_{2f}$, substituting values, we get

$$\frac{T_{1f}}{T_{2f}} = \left(\frac{T_{1i}}{T_{2i}}\right)^{1/\gamma} = \left(\frac{550\text{ K}}{250\text{ K}}\right)^{1/1.4} = 1.756 \quad \textbf{[2]}$$

Solving equations [1] and [2] simultaneously gives

$\boxed{T_{1f} = 510\text{ K}, T_{2f} = 290\text{ K}}$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P21.2 (a) 8.76×10^{-21} J; (b) For helium, $v_{\text{rms}} = 1.62$ km/s and for argon, $v_{\text{rms}} = 514$ m/s

P21.4 3.21×10^{21} molecules

P121.6 $\dfrac{3}{2}\dfrac{PV}{\bar{K}N_A}$

P21.8 2.78×10^{-23} kg · m/s

P21.10 501 K

P21.12 (a) 385 K; (b) 7.97×10^{-21} J; (c) the molecular mass of the gas

P21.14 (a) $W = 0$; (b) $\Delta E_{\text{int}} = 209$ J; (c) 317 K

P21.16 (a) 118 kJ; (b) 6.03×10^3 kg

P21.18 (a) 0.719 kJ/kg · K; (b) 0.811 kg; (c) 233 kJ; (d) 327 kJ

P21.20 Between 10^{-3} °C and 10^{-2} °C

P21.22 (a) See P21.22(a) for full explanation; (b) See P21.22(b) for full explanation; (c) See P21.22(c) for full explanation

P21.24 The maximum possible value of $\gamma = 1 + \dfrac{R}{C_V} = 1.67$ occurs for the lowest possible value for $C_V = \frac{3}{2}R$. Therefore the claim of $\gamma = 1.75$ for the newly discovered gas cannot be true.

P21.26 (a) 1.39 atm; (b) 366 K and 253 K; (c) $Q = 0$; (d) −4.66 kJ; (e) −4.66 kJ

P21.28 (a) 28.0 kJ; (b) 46.0 kJ; (c) 10.0 atm; (d) 25.1 atm

P21.30 The compressed gas would reach a temperature of 941 K, exceeding the melting point of aluminum. Therefore, the claim of improved efficiency using an engine fabricated out of aluminum cannot be true.

P21.32 (a) 2.45×10^{-4} m^3; (b) 9.97×10^{-3} mol; (c) 9.01×10^5 Pa; (d) 5.15×10^{-5} m^3; (e) 560 K; (f) 53.9 J; (g) 6.79×10^{-6} m^3 ; (h) 53.3 g; (i) 2.24 K

P21.34 (a) See ANS. FIG. P21.34(a); (b) $\left(3^{1/\gamma}\right)V_i$; (c) $3T_i$; (d) T_i;

(e) $-P_iV_i\left[\left(\dfrac{1}{\gamma - 1}\right)\left(1 - 3^{1/\gamma}\right) + \left(1 - 3^{1/\gamma}\right)\right]$

P21.36 (a) 6.80 m/s; (b) 7.41 m/s; (c) 7.00 m/s

P21.38 (a) 1.03; (b) ^{35}Cl

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P21.40 (a) 731 m/s; (b) 825 m/s; (c) 895 m/s; (d) The graph appears to be drawn correctly within about 10 m/s.

P21.42 See P21.42 for the full explanation.

P21.44 (a) 3.90 km/s; (b) 4.18 km/s

P21.46 (a) 7.89×10^{26} molecules; (b) 37.9 kg; (c) 6.07×10^{-21} J; (d) 503 m/s; (e) 0; (f) When the furnace operates, air expands and some of it leaves the room. The smaller mass of warmer air left in the room contains the same internal energy as the cooler air initially in the room.

P21.48 (a) 2.26×10^{-9} m; (b) 5.09×10^{-12} seconds

P21.50 (a) See P21.50(a) for the full explanation; (b) 447 J/kg · °C. This agrees with the tabulated value of 448 J/kg · °C within 0.3%; (c) 127 J/kg · °C. This agrees with the tabulated value of 129 J/kg · °C within 2%

P21.52 (a) pressure increases as volume decreases; (b) See P21.52(b) for full answer; (c) See P21.52(c) for full answer; (d) 0.500 atm^{-1}; (e) 0.300 atm^{-1}

P21.54 Sulfur dioxide is the gas with the greatest molecular mass of those listed. If the effective spring constants for various chemical bonds are comparable, SO_2 can then be expected to have low frequencies of atomic vibration. Vibration can be excited at lower temperature than for other gases. Some vibration may be going on at 300 K. With more degrees of freedom for molecular motion, the material has higher specific heat.

P21.56 (a) See P21.56(a) for full explanation; (b) This agrees within 0.2% with the 343 m/s listed in the Table 17.1; (c) See P21.56(c) for full answer; (d) The speed of sound is somewhat less than each measure of molecular speed. Sound propagation is orderly motion overlaid on the disorder of molecular motion.

P21.58 (a) See P21.58(a) for full explanation; (b) See P21.58(b) for full explanation; (c) The expressions are equal because $PV = nRT$ and $\gamma = (C_V + R)/C_V = 1 + R/C_V$ give $R = (\gamma - 1)C_V$, so $PV = n(\gamma - 1)C_V T$ and $PV/(\gamma - 1) = nC_V T$.

P21.60 (a) $T_i + \dfrac{2}{3}\dfrac{W}{nR}$; (b) $P_i\left(1 + \dfrac{2}{3}\dfrac{W}{nRT_i}\right)^{5/2}$

P21.62 (a) See ANS. FIG. P21.62(a); (b) $v_{\text{mp}} \approx 510 \text{ m/s}$; (c) 575 m/s, 624 m/s; (d) 44%

P21.64 (a) 7.27×10^{-20} J/molecule; (b) 2.20 km/s; (c) 3.51×10^3 K; (d) The evaporating particles emerge with much less kinetic energy, as negative work is performed on them by restraining forces as they leave

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the liquid. Much of the initial kinetic energy is used up in overcoming the latent heat of vaporization. There are also very few of these escaping at any moment in time.

P21.66 (a) The larger-mass molecules settle to the outside; (b) $n = n_0 e^{m_0 r^2 \omega^2 / 2k_B T}$

P21.68 (a) $\frac{5}{2}R$; (b) 3R; (c) $\frac{9}{2}R$; (d) 5R; (e) Measure the constant-volume specific heat of the gas as a function of temperature and look for plateaus on the graph. If the first jump goes from $\frac{3}{2}R$ to $\frac{5}{2}R$, the molecules can be diagnosed as linear. If the first jump goes from $\frac{3}{2}R$ to 3R, the molecules must be nonlinear. The tabulated data at one temperature are insufficient for the determination. At room temperature some of the heavier molecules appear to be vibrating.

P21.70 See P21.70 for full explanation.

P21.72 (a) $m_o g R_E$; (b) 1.60×10^4 K

P21.74 (a) See P21.74(a) for full explanation; (b) See P21.74(b) for full explanation; (c) See ANS FIG P21. 74(c); (d) $a = 1$

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